Supplementary Material to "Anatomy of the Treasury Market"

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This Supplementary Material complements the Online Appendix; references to Appendices A and B are to the Online Appendix. Appendix S1 decomposes Treasury yields by factor, Appendix S2 collects further methodological results, Appendix S3 reports additional empirical exhibits, and Appendix S4 contains the proofs.

S1 Yield Decomposition by Factors

This appendix decomposes Treasury yield fluctuations by factor, using the regional elasticity specification in Table 2, panel (b). Appendix B.3.2 reports the decomposition by sector, which follows equation (2.4) of the main text with the regime-specific parameters of equation (3.1); hats denote estimates, $r(t)$ is the regime of quarter t, and $\hat{\zeta}_{S,t} = \sum_i S_{i,t}\hat{\zeta}_{i,r(t)}$ is the quarterly aggregate elasticity.

Each sector's contribution combines the effects of several demand shifters. The factor decomposition separates them: it attributes yield changes to the economic factors that shift demand, such as risk sentiment, inflation, and monetary policy, and identifies the sectors that transmit each factor to prices. Like a reduced-form regression of prices on observed macroeconomic variables, it shows which factors drive Treasury yields; unlike such a regression, it also shows which sectors transmit them.

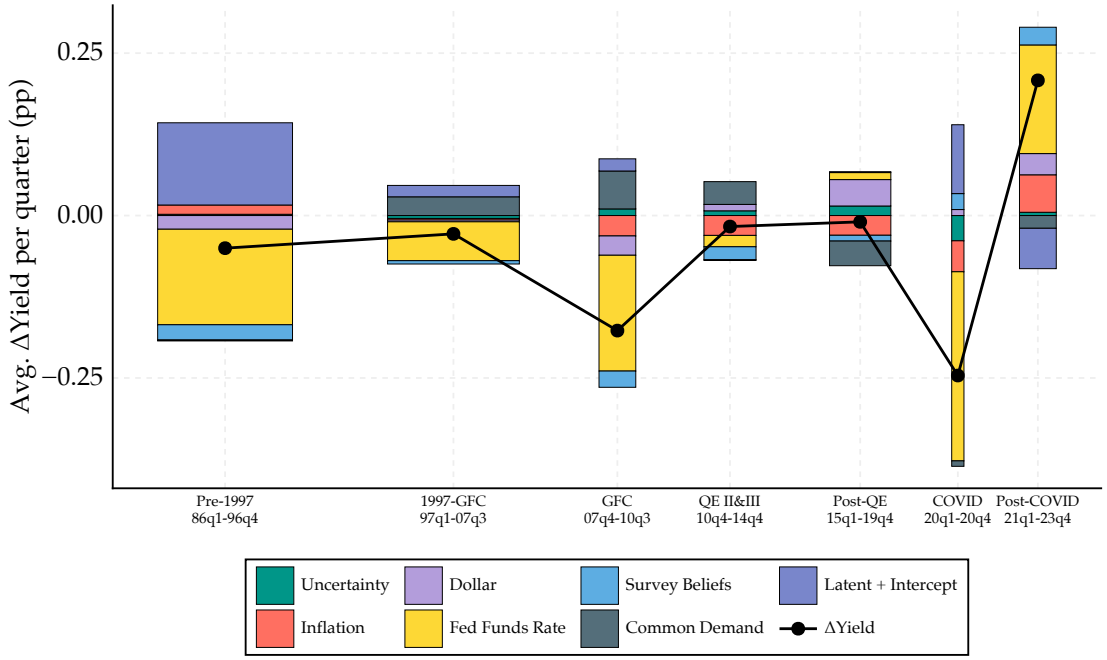
Formally, the contribution of the k-th factor to yield changes is given by:

$$\Delta\hat{p}_{\eta(k),t} = \hat{\zeta}_{S,t}^{-1}\hat{\lambda}_{S,t}(k)\eta_t(k), \tag{S1.1}$$

where $\hat{\lambda}_{S,t}(k) = \sum_i S_{i,t}\hat{\lambda}_{i,r(t)}(k)$ is the quarterly size-weighted sum of factor loadings across sectors. The contribution from idiosyncratic shocks equals $\hat{\zeta}_{S,t}^{-1}\hat{u}_{S,t}$, with $\hat{u}_{S,t} = \sum_i S_{i,t}\hat{u}_{i,t}$. Figure S1.1 combines this contribution with fitted intercepts.

Figure S1.1 illustrates the contribution of various factors to Treasury yield movements, expressed in quarterly averages (percentage points). Table S1.1 reports the full set of average factor loading estimates by sector. We distinguish uncertainty innovations, inflation innovations, U.S. dollar innovations, Federal Funds rate revisions, and four survey forecast factors. The survey factors appear as "Survey Beliefs." Scheduled Federal Reserve (Fed) purchases and lagged net issuance appear as "Common Demand." The decomposition includes sector-specific intercepts, grouped with the idiosyncratic shocks.[1]

Figure S1.1: **Macro and idiosyncratic drivers of Treasury yields**



Note: Contributions are computed from equation (S1.1) using quarterly size weights and the quarterly aggregate price elasticity, then converted to yield changes using lagged Treasury duration. Bars report average contributions per quarter in percentage points; their widths reflect episode lengths, and GFC denotes the global financial crisis. "Survey Beliefs" combines the four principal components of survey forecast revisions; "Common Demand" combines scheduled Fed purchases and lagged net issuance. "Latent + Intercept" combines idiosyncratic shocks and fitted intercepts. All components sum to the observed yield change, shown in black.

The uncertainty component contributes a cumulative yield decline of 15.6 basis points during 2020, but an increase of 12.2 basis points during 2007Q4–2010Q3. This component uses the direct uncertainty loading in the regional specification, with one regime shift at 2009Q1. Section 5 studies uncertainty responses with separate 1997Q3 and 2009Q1 regimes and includes transmission through co-moving macro factors.

[1]Changing the benchmark levels of the factors redistributes contributions between the factors and intercepts while preserving their total.

The Federal Funds rate component contributes a cumulative yield increase of 56.2 basis points during 2004–2006, below one percentage point despite the substantial rise in short rates, consistent with the interest-rate "conundrum" discussed by Greenspan (2005). It contributes declines of 213.9 and 116.4 basis points during the global financial crisis and 2020, respectively, and an increase of 1.98 percentage points during 2022–2023.

Hanson and Stein (2015) emphasize a demand channel through which monetary policy affects long-term rates. Our framework traces the Federal Funds rate component to investor demand loadings. Banks' average loading is negative, associating positive rate revisions with lower Treasury demand, holding Treasury prices and other factors fixed. This sign is consistent with the deposit channel of monetary policy described by Drechsler, Savov, and Schnabl (2017).

CPI inflation innovations contribute a cumulative yield increase of 0.69 percentage points during 2021–2023 and a decline of 37.7 basis points during the global financial crisis. These contributions reflect inflation innovations, rather than inflation expectations. Households' average inflation loading is negative. The negative loading is consistent with households reducing nominal Treasury demand when inflation rises, as in Nagel and Yan (2022).[2]

[2]The Financial Accounts data do not distinguish between nominal Treasury securities and inflation-indexed Treasury bonds (TIPS), limiting our inflation hedging analysis to a high level. However, the TIPS market represents less than 10% of all marketable Treasury debt as of 2024 and was even smaller earlier in our sample. Therefore, we primarily attribute household flows to nominal bonds and notes.

Table S1.1: **Average sector parameter estimates**

Sector	Holdings share (%)	ζ	$\epsilon^{unc}_{(std.)}$	ϵ^{FFR}	Inf.	$\epsilon^{USD}_{(std.)}$	Lagged f_{supply}	Fed schedule
Aggregate		1.01^{***}	0.07	-3.99^{***}	-0.41	-0.46	-0.13	0.19
		[0.89, 1.14]	[−0.49, 0.63]	[−5.99, −2.00]	[−0.95, 0.14]	[−1.06, 0.14]	[−0.69, 0.44]	[−0.62, 1.00]
Supply†	101.01	0.11^{***}	-0.01	-0.13	0.03	0.04	1.74^{***}	0.15
		[0.06, 0.16]	[−0.18, 0.17]	[−0.72, 0.46]	[−0.13, 0.20]	[−0.14, 0.21]	[1.56, 1.91]	[−0.12, 0.42]
Rest of world	38.62	0.35^{***}	-0.38	0.13	-0.02	-0.54	-1.67^{***}	-0.46
		[0.21, 0.49]	[−1.14, 0.39]	[−2.67, 2.94]	[−0.76, 0.73]	[−1.39, 0.30]	[−2.43, −0.91]	[−1.50, 0.58]
Fed	21.88	1.21^{***}	0.84	-19.54^{***}	0.17	-0.37	0.06	4.17^{***}
		[0.98, 1.43]	[−0.27, 1.95]	[−24.53, −14.55]	[−1.00, 1.34]	[−1.77, 1.03]	[−1.01, 1.12]	[3.36, 4.99]
Other	15.22	0.21^{**}	-0.57	-0.41	0.39	0.05	0.06	-0.04
		[0.03, 0.39]	[−1.44, 0.29]	[−3.42, 2.59]	[−0.42, 1.21]	[−0.87, 0.96]	[−0.80, 0.91]	[−1.30, 1.23]
Pension funds	10.79	0.41^{***}	-0.38	-0.33	-0.01	-1.06^{***}	-0.74^{**}	-0.11
		[0.24, 0.58]	[−1.00, 0.25]	[−2.37, 1.71]	[−0.59, 0.58]	[−1.69, −0.43]	[−1.37, −0.11]	[−1.12, 0.90]
Banks	8.55	1.46^{***}	-0.27	-8.73^{***}	0.04	0.88	-3.08^{***}	0.59
		[1.22, 1.71]	[−1.73, 1.19]	[−13.87, −3.58]	[−1.38, 1.46]	[−0.70, 2.45]	[−4.55, −1.62]	[−1.51, 2.68]
Mutual funds	5.99	0.59^{***}	-0.92	0.01	0.29	0.08	-2.77^{***}	0.52
		[0.28, 0.89]	[−2.10, 0.27]	[−4.03, 4.05]	[−0.82, 1.41]	[−1.14, 1.31]	[−3.96, −1.58]	[−1.29, 2.33]
Households†	5.56	0.35^{***}	0.26^{*}	-0.69	-0.37^{**}	-0.23	-0.66^{***}	-0.18
		[0.23, 0.47]	[−0.04, 0.57]	[−1.75, 0.36]	[−0.66, −0.08]	[−0.55, 0.08]	[−0.97, −0.36]	[−0.65, 0.29]
Insurance	5.46	0.44^{***}	-0.48^{*}	-1.35	0.30	-0.42	-1.75^{***}	-0.29
		[0.27, 0.62]	[−1.01, 0.05]	[−3.18, 0.49]	[−0.20, 0.80]	[−0.97, 0.12]	[−2.28, −1.22]	[−1.10, 0.52]
ETFs	1.13	-0.20	1.23	2.16	-0.24	0.91	-1.00	-0.22
		[−0.54, 0.13]	[−0.30, 2.76]	[−4.75, 9.07]	[−1.85, 1.38]	[−1.02, 2.84]	[−2.47, 0.47]	[−1.34, 0.91]
Dealers†	1.07	0.06^{**}	0.13	-0.74^{**}	-0.14	0.06	0.02	-0.27^{*}
		[0.00, 0.12]	[−0.06, 0.32]	[−1.39, −0.10]	[−0.32, 0.04]	[−0.14, 0.25]	[−0.17, 0.21]	[−0.57, 0.02]

Note: This table reports price elasticities ζ_i and direct factor loadings from the specification of Table 1, averaged over each sector's observed quarters in 1986Q1–2023Q4; the rows for the Fed and ETFs cover 2009Q1–2023Q4. The columns are the observed factors of Appendix B.1.2: uncertainty ($\epsilon^{unc}_{(std.)}$), short-rate expectations (ϵ^{FFR}), inflation, the dollar ($\epsilon^{USD}_{(std.)}$), lagged Treasury supply, and scheduled Federal Reserve purchases; the survey-factor loadings are not shown. Holdings share is each sector's average share of Treasury holdings over the same quarters. Sectors marked † have unit size weights, so their coefficients measure demand responses relative to the total market; the other sectors' coefficients are relative to their own holdings. 95% confidence intervals, from equations (A.5) and (A.9), are in brackets below estimates. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

S2 Further Methodological Results

S2.1 Internal Common Factor Estimation

Appendix A.2 treats common factors that are observed or extracted outside the model. In this section we study identification when common demand shocks are unobservable and are estimated together with the other model parameters.

Factor structure for residuals. Suppose the residual $u_{i,t}$ contains common factors $\boldsymbol{\eta}_t$ and idiosyncratic shocks, with $\boldsymbol{\epsilon}_t := (\epsilon_{1,t}, \ldots, \epsilon_{N,t})'$:

$$u_{i,t} := \boldsymbol{\lambda}_i' \boldsymbol{\eta}_t + \epsilon_{i,t}$$

where

$$\mathbb{E}[\boldsymbol{\eta}_t] = \mathbf{0},\ \mathbb{E}[\boldsymbol{\eta}_t \boldsymbol{\eta}_t'] = \mathbf{I}_K,\ \mathbb{E}[\boldsymbol{\epsilon}_t] = \mathbf{0},\ \mathbb{E}[\boldsymbol{\epsilon}_t \boldsymbol{\epsilon}_t'] = \boldsymbol{\Sigma}_\epsilon = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_N^2),\ \mathbb{E}[\boldsymbol{\eta}_t \boldsymbol{\epsilon}_t'] = \mathbf{0}. \quad \text{(S2.1)}$$

We normalize the factors to have unit variance. We also denote $\boldsymbol{\Lambda}$ as the $N \times K$ matrix formed by stacking $\boldsymbol{\lambda}_i'$ row-wise.

Let $\hat{u}_{i,t}(\boldsymbol{\zeta}) = q_{i,t} + p_t \, \mathbf{C}_{i,t}' \boldsymbol{\zeta}$ denote the residual for a candidate elasticity vector $\boldsymbol{\zeta}$. In this case the moment condition for pair (i, j) has an additional term due to the unobserved common factors:

$$\psi_{ij,t}(\boldsymbol{\zeta}; \boldsymbol{\Lambda}) \equiv \hat{u}_{i,t}(\boldsymbol{\zeta}) \, \hat{u}_{j,t}(\boldsymbol{\zeta}) - \boldsymbol{\lambda}_i' \boldsymbol{\lambda}_j, \qquad i \neq j. \quad \text{(S2.2)}$$

Stacking over (i, j) yields $\frac{N(N-1)}{2}$ moments. We can use the generalized method of moments (GMM) estimator for $\boldsymbol{\zeta}$ as in Appendix A.1, replacing the zero off-diagonal restriction with (S2.2) to capture the additional unobserved common variation.

Weighting. With latent factors, $\operatorname{Cov}\big(\psi_{ij,t}, \psi_{kl,t}\big)$ are generally non-zero because of common factors $\boldsymbol{\eta}_t$. Hence, the covariance matrix of the moments V^ψ is no longer diagonal and hence the existing weighting is no longer optimal. In principle, V^ψ can be represented using low-rank matrices involving $\boldsymbol{\Lambda}$, and hence we can still derive the optimal weighting matrix using these representations. In practice, when unobserved factor-induced comovement is relatively small, the diagonal weighting from Appendix A.1 remains close to optimal and efficient for computation, so we adopt it in our baseline implementation. Note that due to off-diagonals being non-zero, the formula for standard errors no longer applies; to address this, we compute bootstrap standard errors.

Unknown loadings: profiled internal PCA. We treat $\mathbf{\Lambda}$ as a nuisance parameter and profile it out i.e. solve for it in the inner loop. For each trial $\boldsymbol{\zeta}$, compute residuals $\hat{u}_{i,t}(\boldsymbol{\zeta})$ and the sample covariance $\widehat{\mathbf{\Sigma}}_{uu}(\boldsymbol{\zeta}) = T^{-1}\sum_t \hat{\mathbf{u}}_t(\boldsymbol{\zeta})\,\hat{\mathbf{u}}_t(\boldsymbol{\zeta})'$. We estimate the factor loadings $\widehat{\mathbf{\Lambda}}(\boldsymbol{\zeta})$ using heteroskedasticity-robust PCA methods (e.g., HeteroPCA or deflated HeteroPCA).

The off-diagonal moment restrictions become

$$\widehat{\psi}_{ij}(\boldsymbol{\zeta}) = \hat{u}_{i,t}(\boldsymbol{\zeta})\,\hat{u}_{j,t}(\boldsymbol{\zeta}) - \hat{\boldsymbol{\lambda}}_i(\boldsymbol{\zeta})'\,\hat{\boldsymbol{\lambda}}_j(\boldsymbol{\zeta}), \qquad i \neq j, \tag{S2.3}$$

S2.1.1 Identifiability

When we estimate $\mathbf{\Lambda}$ internally, uniqueness of the solution is no longer guaranteed and depends on the exact specification. As a rule of thumb, we show that identification fails when $N_\zeta + K > N$. Below we provide precise conditions that clarify when the moment condition has a non-unique solution.

Moment representation. Using matrix notation, we define the observable second moments as:

$$\mathbf{\Sigma}_{qq} := \mathbb{E}[\mathbf{q}_t\mathbf{q}_t'] \in \mathbb{R}^{N\times N}, \qquad \mathbf{\Sigma}_{qp} := \mathbb{E}[\mathbf{q}_t p_t] \in \mathbb{R}^{N\times 1}, \qquad \sigma_p^2 := \mathbb{E}[p_t^2] \in \mathbb{R},$$

For tractability, in this section we consider the elasticity loading matrix $\mathbf{C}$ to be time-invariant. The moment conditions are then given in terms of observables and parameters:

$$\mathbf{\Sigma}_{qq} \;+\; \mathbf{\Sigma}_{qp}\,(\mathbf{C}\boldsymbol{\zeta})' \;+\; (\mathbf{C}\boldsymbol{\zeta})\,\mathbf{\Sigma}_{qp}' \;+\; \sigma_p^2\,(\mathbf{C}\boldsymbol{\zeta})(\mathbf{C}\boldsymbol{\zeta})' = \mathbf{\Lambda}\,\mathbf{\Lambda}' \;+\; \mathbf{\Sigma}_\epsilon. \tag{S2.4}$$

We denote $(\boldsymbol{\zeta}_\star, \mathbf{\Lambda}_\star, \mathbf{\Sigma}_\star)$ as the true parameters. Identification means $(\boldsymbol{\zeta}_\star, \mathbf{\Lambda}_\star, \mathbf{\Sigma}_\star)$ is the unique solution to the moment condition (S2.4). Equivalently, let $(\tilde{\boldsymbol{\zeta}}, \tilde{\mathbf{\Lambda}}, \tilde{\mathbf{\Sigma}})$ denote deviations from the truth.Identification requires that the only solution to the perturbed system (S2.5) is $\tilde{\boldsymbol{\zeta}} = \mathbf{0}$, $\tilde{\mathbf{\Lambda}} = \mathbf{0}$, and $\tilde{\mathbf{\Sigma}} = \mathbf{0}$.

$$(\mathbf{C}\,\tilde{\boldsymbol{\zeta}})\,\mathbf{\Sigma}_{qp}' + \mathbf{\Sigma}_{qp}\,(\mathbf{C}\,\tilde{\boldsymbol{\zeta}})' + \sigma_p^2\Big((\mathbf{C}\,\tilde{\boldsymbol{\zeta}})(\mathbf{C}\,\boldsymbol{\zeta}_\star)' + (\mathbf{C}\,\boldsymbol{\zeta}_\star)(\mathbf{C}\,\tilde{\boldsymbol{\zeta}})' + (\mathbf{C}\,\tilde{\boldsymbol{\zeta}})(\mathbf{C}\,\tilde{\boldsymbol{\zeta}})'\Big) = \mathbf{\Lambda}_\star\tilde{\mathbf{\Lambda}}' + \tilde{\mathbf{\Lambda}}\,\mathbf{\Lambda}_\star' + \tilde{\mathbf{\Lambda}}\tilde{\mathbf{\Lambda}}' + \tilde{\mathbf{\Sigma}}. \tag{S2.5}$$

To characterize identification more clearly, we study local identifiability around the true parameters by focusing on the first-order approximation to (S2.5). Dropping the

quadratic terms and refactoring, the first-order approximation is

$$(\mathbf{C}\,\tilde{\boldsymbol{\zeta}})\left(\boldsymbol{\Sigma}_{qp}+\sigma_p^2\,\mathbf{C}\,\boldsymbol{\zeta}_\star\right)' + \left(\boldsymbol{\Sigma}_{qp}+\sigma_p^2\,\mathbf{C}\,\boldsymbol{\zeta}_\star\right)(\mathbf{C}\,\tilde{\boldsymbol{\zeta}})' - \tilde{\boldsymbol{\Sigma}} = \boldsymbol{\Lambda}_\star\,\tilde{\boldsymbol{\Lambda}}' + \tilde{\boldsymbol{\Lambda}}\,\boldsymbol{\Lambda}_\star'. \tag{S2.6}$$

Proposition S2.1. *The parameters are first-order identified if and only if the only solution to the following equation is* $\tilde{\boldsymbol{\zeta}} = \mathbf{0}$*:*

$$\mathbf{Q}^\Lambda\,\mathbf{C}\,\tilde{\boldsymbol{\zeta}}\left(\mathbf{S}'\boldsymbol{\Sigma}_\star\right)\mathbf{Q}^\Lambda + \mathbf{Q}^\Lambda\left(\boldsymbol{\Sigma}_\star\,\mathbf{S}\right)\tilde{\boldsymbol{\zeta}}'\mathbf{C}'\,\mathbf{Q}^\Lambda = \zeta_S^\star\,\mathbf{Q}^\Lambda\,\tilde{\boldsymbol{\Sigma}}\,\mathbf{Q}^\Lambda, \tag{S2.7}$$

where $\mathbf{R}^\Lambda := \boldsymbol{\Lambda}_\star(\boldsymbol{\Lambda}_\star'\boldsymbol{\Lambda}_\star)^{-1}\boldsymbol{\Lambda}_\star'$*, and* $\mathbf{Q}^\Lambda := \mathbf{I} - \mathbf{R}^\Lambda$ *as the null space projector with respect to* $\boldsymbol{\Lambda}_\star$*, so that* $\mathbf{R}^\Lambda\boldsymbol{\Lambda}_\star = \boldsymbol{\Lambda}_\star$ *and* $\mathbf{Q}^\Lambda\boldsymbol{\Lambda}_\star = \mathbf{0}$*.*

Proof. Necessity: Expressing $\boldsymbol{\Sigma}_{qp}$ and σ_p^2 in terms of the parameters, we have

$$\boldsymbol{\Sigma}_{qp} + \sigma_p^2\,\mathbf{C}\,\boldsymbol{\zeta}_\star = \frac{\boldsymbol{\Lambda}_\star\boldsymbol{\Lambda}_{S,\star}' + \boldsymbol{\Sigma}_\star\,\mathbf{S}}{\zeta_S^\star}, \quad \text{where } \boldsymbol{\Lambda}_{S,\star} := \mathbf{S}'\boldsymbol{\Lambda}_\star,\ \zeta_S^\star := \mathbf{S}'\mathbf{C}\,\boldsymbol{\zeta}_\star.$$

Multiplying (S2.6) by $\mathbf{Q}^\Lambda$ on both sides and simplifying using $\mathbf{Q}^\Lambda\boldsymbol{\Lambda}_\star = \mathbf{0}$ yields (S2.7).

Sufficiency: Suppose (S2.7) holds for a nonzero $\tilde{\boldsymbol{\zeta}}$ and a diagonal $\tilde{\boldsymbol{\Sigma}}$. Define the left-hand side of (S2.6) as

$$\mathbf{L} := (\mathbf{C}\,\tilde{\boldsymbol{\zeta}})\left(\boldsymbol{\Sigma}_{qp}+\sigma_p^2\,\mathbf{C}\,\boldsymbol{\zeta}_\star\right)' + \left(\boldsymbol{\Sigma}_{qp}+\sigma_p^2\,\mathbf{C}\,\boldsymbol{\zeta}_\star\right)(\mathbf{C}\,\tilde{\boldsymbol{\zeta}})' - \tilde{\boldsymbol{\Sigma}}.$$

By (S2.7) and $\mathbf{Q}^\Lambda\boldsymbol{\Lambda}_\star = \mathbf{0}$, we have $\mathbf{Q}^\Lambda\,\mathbf{L}\,\mathbf{Q}^\Lambda = \mathbf{0}$. Now set

$$\tilde{\boldsymbol{\Lambda}} := \mathbf{L}\,\boldsymbol{\Lambda}_\star\,(\boldsymbol{\Lambda}_\star'\boldsymbol{\Lambda}_\star)^{-1} - \tfrac{1}{2}\,\mathbf{R}^\Lambda\,\mathbf{L}\,\boldsymbol{\Lambda}_\star\,(\boldsymbol{\Lambda}_\star'\boldsymbol{\Lambda}_\star)^{-1}.$$

Using $\mathbf{L} = \mathbf{L}'$ and $(\mathbf{R}^\Lambda)' = \mathbf{R}^\Lambda$, we compute

$$\boldsymbol{\Lambda}_\star\,\tilde{\boldsymbol{\Lambda}}' = \mathbf{R}^\Lambda\,\mathbf{L} - \tfrac{1}{2}\,\mathbf{R}^\Lambda\,\mathbf{L}\,\mathbf{R}^\Lambda, \qquad \tilde{\boldsymbol{\Lambda}}\,\boldsymbol{\Lambda}_\star' = \mathbf{L}\,\mathbf{R}^\Lambda - \tfrac{1}{2}\,\mathbf{R}^\Lambda\,\mathbf{L}\,\mathbf{R}^\Lambda.$$

Adding yields

$$\boldsymbol{\Lambda}_\star\,\tilde{\boldsymbol{\Lambda}}' + \tilde{\boldsymbol{\Lambda}}\,\boldsymbol{\Lambda}_\star' = (\mathbf{R}^\Lambda\,\mathbf{L} + \mathbf{L}\,\mathbf{R}^\Lambda) - \mathbf{R}^\Lambda\,\mathbf{L}\,\mathbf{R}^\Lambda.$$

Using the decomposition $\mathbf{L} = (\mathbf{R}^\Lambda + \mathbf{Q}^\Lambda)\,\mathbf{L}\,(\mathbf{R}^\Lambda + \mathbf{Q}^\Lambda)$, the right-hand side equals $\mathbf{L} - \mathbf{Q}^\Lambda\,\mathbf{L}\,\mathbf{Q}^\Lambda = \mathbf{L}$, because $\mathbf{Q}^\Lambda\,\mathbf{L}\,\mathbf{Q}^\Lambda = \mathbf{0}$. Therefore, (S2.6) holds, proving lack of first-order identification. □

The general characterization of identification depends on the exact $\mathbf{C}$. However,

corollaries below offer two sufficient conditions when the first-order identification fails.

Corollary 1. *The parameters are not first-order identified when* $N_\zeta + K > N$.

Proof. When $N_\zeta + K > N$, we have $\mathcal{S} := \mathrm{col}(\mathbf{C}) \cap \mathrm{col}(\mathbf{\Lambda}_\star) \neq \{\mathbf{0}\}$. Pick any non-zero $\mathbf{v} \in \mathcal{S}$ and choose $\tilde{\boldsymbol{\zeta}}$ such that $\mathbf{C}\,\tilde{\boldsymbol{\zeta}} = \mathbf{v}$. Since $\mathbf{v} \in \mathrm{col}(\mathbf{\Lambda}_\star)$, we have $\mathbf{Q}^{\Lambda}\mathbf{C}\,\tilde{\boldsymbol{\zeta}} = \mathbf{Q}^{\Lambda}\mathbf{v} = \mathbf{0}$. With $\tilde{\mathbf{\Sigma}} = \mathbf{0}$, Equation (S2.7) holds. □

The corollary states that the number of elasticity coefficients plus the number of factors is greater than the number of entities. Therefore, with fully heterogeneous elasticities we cannot identify factors internally. Empirically, identification is restored by imposing structure—e.g., grouping entities to share elasticities or restricting K—so that $\mathrm{col}(\mathbf{C})$ is sufficiently "transverse" to $\mathrm{col}(\mathbf{\Lambda}_\star)$ and the dimension count is favorable.

Corollary 2. *The parameters are not first-order identified when* $\mathbf{\Sigma}_\star\mathbf{S} \in \mathrm{col}(\mathbf{\Lambda}_\star)$.

Proof. When $\mathbf{\Sigma}_\star\mathbf{S} \in \mathrm{col}(\mathbf{\Lambda}_\star)$, we have $\mathbf{Q}^{\Lambda}\mathbf{\Sigma}_\star\mathbf{S} = \mathbf{0}$, hence Equation (S2.7) holds for any $\tilde{\boldsymbol{\zeta}}$. □

This is akin to the granularity condition required by Gabaix and Koijen (2024) in the original GIV setup—the condition is satisfied when the size–volatility distribution is uniform, so $\mathbf{\Sigma}_\star\mathbf{S} \propto \mathbf{1}$ and there exists a common factor with homogeneous loading. Intuitively, to disentangle price elasticities from unobserved common factors, we rely on investors' influence on the price $S_i\,\sigma_i^2$—larger and more volatile investors have larger price impacts. However, when their influence is perfectly aligned with the loadings on the common factors, we can no longer distinguish elasticity from factor loadings.

S2.2 Comparison with the Standard GIV

The size weighting in the optimal granular instrumental variables (GIV) estimator resembles the original GIV method of Gabaix and Koijen (2024). The two estimators are closely linked but differ in ways that matter for statistical power. In this section we discuss the relationship of our approach to the method introduced in Gabaix and Koijen (2024), hereafter referred to as the standard GIV.

On top of the assumptions in Appendix A.1, the standard GIV further imposes homogeneity in elasticities across entities.[3] To closely compare two estimators, we also consider

[3]Gabaix and Koijen (2024) also offer a method to handle heterogeneity in the online appendix, which is further developed by Chodorow-Reich, Gabaix, Koijen, and Viviano (2024). Here the standard GIV refers to the case with homogeneity introduced in the main text of Gabaix and Koijen (2024).

the case with homogeneous elasticity so that $\zeta_i \equiv \zeta$, and we also consider the case where the size vector sums to 1, $\sum_i S_i = 1$.

With homogeneity, the optimal GIV estimator pools the sample moment conditions (3.3) across i weighted by the inverse of residual variance $(\sigma_i^2)^{-1}$ (see Appendix A.1), so that the optimal GIV estimator ζ^{OGIV} corresponds to the following moment conditions in population:

$$\sum_i \frac{(\sigma_i^2)^{-1}}{\Sigma_i(\sigma_i^2)^{-1}} \mathbb{E}\left[\left(q_{i,t} + \zeta^{OGIV} p_t\right) \sum_{j \neq i} S_j \left(q_{j,t} + \zeta^{OGIV} p_t\right)\right] = 0$$

Denote the $\mathbf{\Sigma}_u = diag\left(\sigma_1^2, \sigma_2^2, \ldots, \sigma_N^2\right)$ as the covariance matrix of the residual variance, and $\mathbf{E}_i \equiv \frac{(\sigma_i^2)^{-1}}{tr(\mathbf{\Sigma}_u^{-1})}$ as the weights using the inverse of the residual volatility, we can consolidate the corresponding moment conditions above in the matrix form:

$$\mathbf{E}'\mathbb{E}\left[\left(\mathbf{q}_t + \boldsymbol{\iota}\zeta^{OGIV} p_t\right)\left(\mathbf{q}_t + \boldsymbol{\iota}\zeta^{OGIV} p_t\right)'\right]\mathbf{S} = \frac{1}{tr(\mathbf{\Sigma}_u^{-1})}. \qquad \text{(Optimal-GIV)}$$

Denote $\mathrm{Off}(\mathbf{A})$ as the matrix that sets the diagonal elements of $\mathbf{A}$ to zero, we can also express the moment conditions as:

$$\mathbf{E}' \,\mathrm{Off}\left(\mathbb{E}\left[\left(\mathbf{q}_t + \boldsymbol{\iota}\zeta^{OGIV} p_t\right)\left(\mathbf{q}_t + \boldsymbol{\iota}\zeta^{OGIV} p_t\right)'\right]\right)\mathbf{S} = 0.$$

The estimating equation is quadratic in ζ because the moment condition multiplies two terms that each depend linearly on ζ, so we solve for ζ^{OGIV} numerically.

The key insight in the standard GIV is that, with homogeneous elasticities, cross-sectional demeaning can eliminate the endogenous term ζp_t. Mathematically, for any vector that sums to 1, $\mathbf{X}'\boldsymbol{\iota} = 1$, we also have $\mathbf{E}'\mathbf{\Sigma}_u\mathbf{X} = \frac{1}{tr(\mathbf{\Sigma}_u^{-1})}$, so by taking the difference of these two equations, we have:

$$\mathbf{E}'\mathbb{E}\left[\left(\mathbf{q}_t + \boldsymbol{\iota}\zeta^{SGIV} p_t\right)\left(\mathbf{q}_t + \boldsymbol{\iota}\zeta^{SGIV} p_t\right)'\right](\mathbf{S} - \mathbf{X}) = 0, \qquad \text{(Standard-GIV)}$$

where $\mathbf{S} - \mathbf{X}$ essentially demeans $\mathbf{q}_t$ in the cross section. Move weighting vectors into the expectation operator, this weighting scheme can be further expressed as in terms of weighted average of q:

$$\mathbb{E}\left[\left(q_{E,t} + \zeta^{SGIV} p_t\right)\left(q_{S,t} - q_{X,t}\right)\right] = 0.$$

Importantly, the term ζp_t drops out from the second term in the expectation due to differencing. As it is linear in ζ, it can now be implemented with an IV regression of precision-weighted quantity $(q_{E,t})$ on p_t, using $q_{S,t} - q_{X,t}$ as the instrument. Gabaix and Koijen (2024) further show that within this class of the linear estimators, setting $\mathbf{X} = \mathbf{E}$ achieves the highest statistical power, and hence the recommended instrument is the size-minus-precision weighted q.

At first glance, the mathematical difference between the standard GIV and the optimal GIV in the simplest case is merely whether to demean $q_{i,t}$ in the cross section. This difference nevertheless matters for statistical power in a sample where the distribution of the contribution to price changes across entities does not exhibit a large fat tail.

To see the power comparison, consider a simpler case with homoskedasticity, i.e., $\sigma_i \equiv \sigma$. In this case, the asymptotic variance of the optimal GIV estimator has a simple expression: [4]

$$\sqrt{T}\left(\hat{\zeta}^{OGIV} - \zeta\right) \xrightarrow{d} \mathcal{N}\left(0, \mathbf{V}_\zeta^{OGIV} = \frac{\zeta^2}{(N-2)\sum_i S_i^2 + 1}\right),$$

while the standard GIV converges to a different limiting distribution: [5]

$$\sqrt{T}\left(\hat{\zeta}^{SGIV} - \zeta\right) \xrightarrow{d} \mathcal{N}\left(0, \mathbf{V}_\zeta^{SGIV} = \frac{\zeta^2}{N\sum_i S_i^2 - 1}\right).$$

We can show that $\mathbf{V}_\zeta^{OGIV} < \mathbf{V}_\zeta^{SGIV}$, with equality only in the degenerate case where a single entity holds the entire market share. Moreover, the standard GIV relies on the fat-tail distribution in S_i for statistical power, while the optimal GIV does not. This is because when sizes are equally distributed $(S_i = \frac{1}{N})$, the size weighting and precision weighting (which is equal weighting in the case of homoskedasticity) coincide, so the standard GIV's instrument has zero variation after demeaning. This is not the case in the optimal GIV, whose statistical power still grows at the rate of $\sqrt{T}$.

The power comparison matters in practice. While fat-tail distribution in sizes is prevalent in real-world data, the residual volatility typically shrinks as an entity becomes larger. As shown in the expressions in (3.4), what matters for the statistical power is not only the size distribution but the distribution of each entity's contribution to price volatility, i.e., the size-weighted volatility, $S_i^2\sigma_i^2$. To see that, consider the case where the size has a fat tail distribution but the residual volatility is inversely proportional to the size,

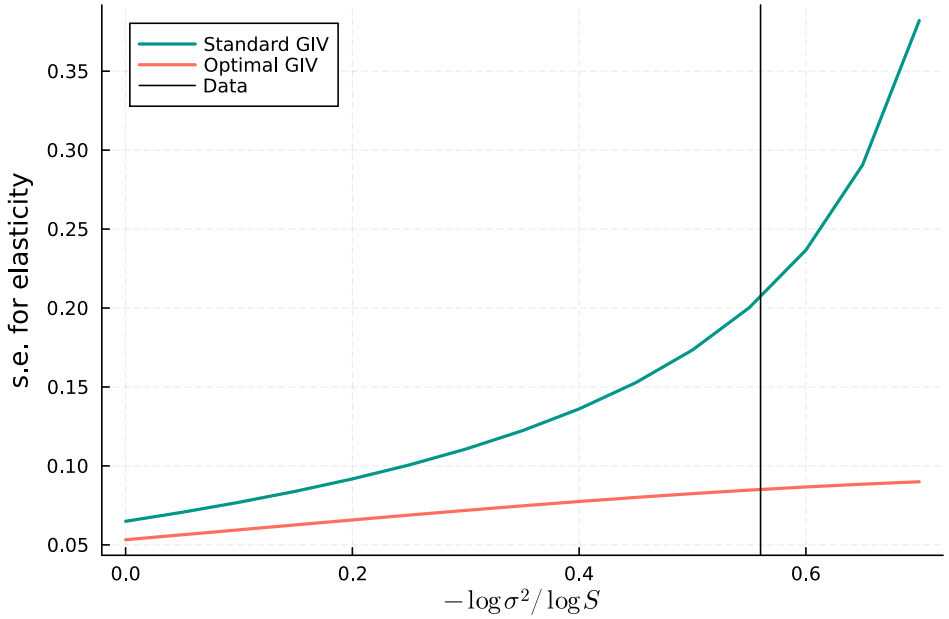
[4] See Appendix S4.1 for the derivation.
[5] See Gabaix and Koijen (2024) for the derivation.

$\sigma_i^2 \propto \frac{1}{S_i}$. In this case, the size vector again coincides with the precision weighting, and there is no variation in the size-minus-precision instruments.

Figure S2.1 compares the statistical power of the standard GIV in Gabaix and Koijen (2024) vs. the optimal GIV in simulation. We simulate the model mirroring the Treasury market in the Financial Accounts data. We assume there are 12 sectors with a homogeneous elasticity of 1, and the supply is inelastic. The size distribution is calibrated with the concentration in the Financial Accounts data, and we vary the rate at which volatility decreases as the size of a sector increases. Other model parameters are reported in the note accompanying Figure S2.1. As the volatility decreases, the precision of standard GIV decays rapidly, while only mildly for the optimal GIV. When the volatility-size ratio matches that in the Financial Accounts data, the optimal GIV is twice as powerful as the standard GIV.

The higher statistical power of the optimal GIV comes with two trade-offs. First, the optimal GIV must be solved numerically, which adds computational complexity. As with any numerical scheme, multiple roots may exist or convergence may fail when the sample size is small. Second, the standard GIV automatically handles common factors with homogeneous loadings by cross-sectional demeaning. This step eliminates potential common factors not controlled for in the data and is, in this sense, more conservative, whereas the optimal GIV by default interprets all residuals as idiosyncratic shocks. Appendix S2.1 describes how to extend the optimal GIV to account for unobserved common factors internally.

Figure S2.1: **Statistical power comparison: standard GIV vs. optimal GIV**



Note: This figure compares the statistical power of the standard GIV in Gabaix and Koijen (2024) vs. the optimal GIV in simulation. Simulation parameters: number of sectors $N = 12$, $T = 80$ quarters, no common factors, the excess HHI in the size distribution, $h \equiv \sqrt{\sum_i S_i^2 - \frac{1}{N}} = 0.5$, elasticity $\zeta = 1.0$, and the level of the shock volatility is calibrated so that the $\sigma_p^2 = 2.5\%$.

S2.3 Structural Interpretations: Derivations

This subsection derives the micro-foundations summarized in Table A.1, using the notation of Appendix A.4.

CRRA portfolio choices. The canonical portfolio choice model features constant relative risk aversion (CRRA). Consider investor i choosing between the short-term risk-free asset and a long-term risky bond. The implied portfolio demand is

$$\frac{Q_{i,t}P_t}{W_{i,t}} = \frac{\mu_t - r_t}{\gamma_{i,t}\sigma_t^2}$$

where $1 + \mu_t \equiv \frac{C+\mathbb{E}_t[P_{t+1}]}{P_t}$ is the one-period expected return on the long-term bond (with C the coupon), r_t is the short-term risk-free return, $\gamma_{i,t}$ is risk aversion, σ_t^2 is the variance of Treasury returns, and $W_{i,t}$ is wealth.

Log-linearizing around long-run levels, with the notation of Appendix A.4, yields

the demand curve:

$$\hat{q}_{i,t} = \underbrace{-\left(1 + \frac{1+\bar{\mu}}{\bar{\mu}-\bar{r}}\right)}_{\zeta_i} \hat{p}_t + \underbrace{\frac{1}{\bar{\mu}-\bar{r}} \widehat{\mathbb{E}_t[p_{t+1}]} + \hat{w}_{i,t} - \frac{1}{\bar{\mu}-\bar{r}} \hat{r}_t - \hat{\gamma}_{i,t} - \hat{\sigma}_t^2}_{\nu_{i,t}}.$$

Hence, in the standard CRRA case, the elasticity is pinned down by the average expected return and the risk-free rate. The demand shifter $\nu_{i,t}$ reflects fluctuations in wealth, the expected future price, the risk-free rate, risk aversion, and return volatility.

As well-known in the literature, a frictionless CRRA model typically implies unrealistically high elasticities (Gabaix & Koijen, 2022). As a numerical example, let the average expected return be $\bar{\mu} = 5\%$ and the risk-free rate $\bar{r} = 2\%$. Then $\zeta_i \approx 1 + \frac{1.05}{0.05-0.02} \approx 36$, far above our estimates. Several extensions can generate empirically plausible elasticities, as discussed below.

Learning-from-price models. Under the CRRA benchmark we assume rational expectations, so expectations of future prices are independent of the current price. However, when investors receive noisy private signals about future prices, the current price aggregates dispersed private information and investors learn from it. A large literature on learning from prices analyzes this behavior; see, for example, Kyle (1985), Hellwig (1980), and Grossman and Stiglitz (1980).

Continuing from (S2.3), suppose investor i receives a private noisy signal $s_{i,t}$ and understands that the current price $\hat{p}_t$ contains information about other investors' private signals. In this case, the investor's expectation of the future price is a linear combination of the private signal and the current price:

$$\widehat{\mathbb{E}_t[p_{t+1}]} = \phi_p \hat{p}_t + \phi_s s_{i,t}$$

The coefficients $\phi_p > 0$ and $\phi_s > 0$ depend on the information precision in the price and in the private signal, respectively, and are pinned down in equilibrium.

Plugging this into (S2.3), the demand under learning from price is

$$\hat{q}_{i,t} = \underbrace{-\left(1 + \frac{1+\bar{\mu}-\phi_p}{\bar{\mu}-\bar{r}}\right)}_{\zeta_i} \hat{p}_t + \underbrace{\frac{\phi_s}{\bar{\mu}-\bar{r}} s_{i,t} + \hat{w}_{i,t} - \frac{1}{\bar{\mu}-\bar{r}} \hat{r}_t - \hat{\gamma}_{i,t} - \hat{\sigma}_t^2}_{\nu_{i,t}}.$$

Under learning from price, investors infer future prices from the current price. When the current price rises, they infer that the market has priced in private signals indicating

higher future prices, so they are less inclined to trade against price increases. Therefore, learning from price (larger ϕ_p) reduces the elasticity. If investors rely heavily on the current price, the demand may not be downward-sloping ($\zeta_i \leq 0$), consistent with passive or momentum trading.

Convenience yield models. Beyond risk–return trade-offs, investors may hold Treasuries for their convenience yield. A large literature documents that, for the same payoff, investors are willing to pay a premium to hold Treasuries because they provide liquidity services and serve as safe assets (Krishnamurthy & Vissing-Jorgensen, 2012; Nagel, 2016; Fu, Li, & Xie, 2022).

Convenience services are often modeled via a bonds-in-utility specification. Consider an investor who derives utility from both consumption and Treasury holdings, which are typically treated as risk-free in this framework. The implied condition (see (Krishnamurthy & Vissing-Jorgensen, 2012) for the full setup and derivation) is

$$\frac{(Q_{i,t}P_t)^{-\psi_i}}{C_{i,t}^{-\theta_i}} = r_t - y_t \equiv cy_t$$

where $y_t \equiv \frac{F}{P_t} - 1$ is the yield of the Treasury bond, $C_{i,t}$ is the consumption, and r_t is the return from a risk-free bond without the convenience service.

Log-linearizing around long-run levels gives

$$\hat{q}_{i,t} = \underbrace{-\left(1 + \frac{1+\bar{y}}{\psi_i \overline{cy}}\right)}_{\zeta_i^{cy}} \hat{p}_t + \underbrace{\frac{\theta_i}{\psi_i}\hat{c}_{i,t} - \frac{1}{\psi_i \overline{cy}}\hat{r}_t}_{\nu_{i,t}^{cy}}.$$

Hence the elasticity is governed by the curvature parameter ψ_i associated with convenience services.

Liability-driven investment models. Liability-driven investment is commonly adopted by insurance companies and pension funds (ICPFs). ICPFs typically have long-duration liabilities and hold long-term bonds to hedge the interest-rate risk of their liability exposures. Suppose ICPFs match the dollar duration of their assets and liabilities; equivalently, they match present value (PV) times modified duration on both sides (Domanski, Shin, & Sushko, 2017):

$$Q_{i,t}P_t D_t^T(P_t) = L_{i,t}(y_t) D_{i,t}^L(P_t)$$

where D_t^T and $D_{i,t}^L$ are modified durations (in years) of Treasuries and liabilities, respectively, and $L_{i,t}$ is the present value (PV) of liabilities. This equates dollar-duration exposures: $Q_{i,t}P_tD_t^T = L_{i,t}D_{i,t}^L$.

Log-linearize around a steady state. Taking logs of $Q_{i,t}P_tD_t^T = L_{i,t}D_{i,t}^L$ (and writing $d_{i,t}^X \equiv \ln D_{i,t}^X$) gives

$$q_{i,t} + p_t + d_t^T = l_{i,t} + d_{i,t}^L \quad \Rightarrow \quad \hat{q}_{i,t} = -\hat{p}_t - \hat{d}_t^T + \hat{l}_{i,t} + \hat{d}_{i,t}^L.$$

Hence the LDI demand is

$$\hat{q}_{i,t} = -\underbrace{\left(1 + \kappa^T - \kappa_i^L\right)}_{:=\zeta_i^{LDI}} \hat{p}_t + \underbrace{\hat{l}_{i,t}}_{\nu_{i,t}^{LDI}}.$$

where we define the elasticities of duration with respect to price as:

$$\kappa^T := \frac{\partial d_t^T}{\partial p_t} = \frac{\partial \log D_t^T}{\partial \log P_t}, \qquad \kappa_i^L := \frac{\partial d_{i,t}^L}{\partial p_t} = \frac{\partial \log D_{i,t}^L}{\partial \log P_t}.$$

Typically, liabilities for ICPFs have much longer duration than Treasury assets and similar dispersion in cash flows, so $\kappa_i^L > \kappa^T$ and hence $\zeta_i^{LDI} < 1$. [6]

[6] To see this, consider price movement due to a parallel shift in the yield curve. Using $\frac{\partial \ln P_t}{\partial y_t} = -D_t^T$ and $\frac{\partial \ln D_t^X}{\partial y_t} = D_t^X - \frac{C_t^X}{D_t^X}$ with respect to the underlying yield y_t (where $C_t^X := \frac{1}{P_t^X}\frac{\partial^2 P_t^X}{\partial y_t^2}$ is convexity and $X \in \{T, L\}$), the chain rule implies

$$\kappa^T = -1 + \frac{C_t^T}{(D_t^T)^2}, \qquad \kappa^L = -\frac{D_t^L}{D_t^T} + \frac{C_t^L}{D_t^L D_t^T} = \frac{D_t^L}{D_t^T}\Big(-1 + \frac{C_t^L}{(D_t^L)^2}\Big).$$

For any cash-flow stream, $C_t^X = E_w[t^2]$ and $D_t^X = E_w[t]$ for the cash-flow time distribution w, so $-1 + C_t^X/(D_t^X)^2 = \mathrm{Var}_w[t]/(E_w[t])^2 \geq 0$. Hence κ is nonnegative and increasing in the dispersion/convexity of cash flows, and tends to be larger for longer-duration portfolios with similar dispersions. In particular, if liabilities typically have longer duration than Treasury assets ($D_t^L > D_t^T$) and comparable dispersion, then $\kappa^L > \kappa^T$.

S2.4 Additional Discussions on Optimal GIV Estimator

S2.4.1 When only a subset of sectors are observed

Consider the case where only a subset of sectors are observed. In this case,

$$\left.\begin{array}{rl} q_{i,t} & = -p_t \times \mathbf{C}'_{i,t}\boldsymbol{\zeta} + \mathbf{X}'_{i,t}\boldsymbol{\beta} + u_{i,t}, \\ q_{i,t}^{unobs} & = -\zeta_t^{unobs} p_t + \epsilon_t \\ 0 & = \sum_i S_{i,t} q_{i,t} + q_t^{unobs} \end{array}\right\} \implies \quad p_t = M_t \left[\mathbf{X}'_{S,t}\boldsymbol{\beta} + u_{S,t} + \epsilon_t\right], \tag{S2.8}$$

where $M_t = \left(\mathbf{C}'_{S,t}\boldsymbol{\zeta} + \zeta_t^{unobs}\right)^{-1}$ is the inverse of the sum of the observed and unobserved sectors' demand elasticities. In this case, whether we can correctly identify the elasticity depends on the covariance structure of the error terms.

Consistency. Notice that to identify ζ_i for observed sectors, the original moment conditions $\mathbb{E}\left[u_{i,t}u_{j,t}\right] = 0$ suffice, even if $u_{i,t}$ can be correlated with the unobserved sector's error term $\epsilon_{i,t}$. Using the language of the IV interpretation, the IV is valid regardless of how the instruments affect the endogenous variable p_t, as long as the instruments affect the outcome variable $q_{i,t}$ only through p_t.

The aggregate elasticity, or the multiplier M_t, cannot be identified simply by $\mathbf{C}'_{S,t}\boldsymbol{\zeta}$, due to the elasticity provision of the unobserved sector. If we have $\mathbb{E}\left[u_{i,t}\epsilon_t \mid \mathbf{X}_{i,t}\right] = 0$, i.e., the demand shifts from the unobserved sectors are orthogonal to idiosyncratic shocks conditional on common factors, we can recover M_t by regressing p_t on $u_{S,t}$. Alternatively, if we can assume that the observed sectors are representative of the unobserved sectors, then we can impute the ζ_t^{unobs} using the observed sectors' $\mathbf{C}'_{i,t}\boldsymbol{\zeta}$.

S2.4.2 Measurement Errors

The quantity data can be noisy in real world applications. The quantity observed by the econometrician may be contaminated by the additional measurement errors:

$$\tilde{q}_{i,t} = q_{i,t} + \varepsilon_{i,t},$$

where $\varepsilon_{i,t}$ only enters the observed $\tilde{q}_{i,t}$ but not the real $q_{i,t}$ and hence not the pricing equation (2.3).

If the measurement errors are *classic*, i.e., it is independent with prices, common factors and each other, then the optimal GIV estimator is still consistent, as the moment conditions in (3.2) still holds with additional orthogonal terms $\varepsilon_{i,t}$. The size-weighting

scheme, however, will no longer be optimal except for the special case where $\mathrm{Var}\left(\varepsilon_{i,t}\right) \propto \mathrm{Var}(u_{i,t})$.

Nevertheless, the existence of a residual sector may break the consistency of the estimator under classic measurement errors. A residual sector is a sector that is not directly observed in the data, but backed out from the market clearing condition. In real world applications, we typically do not observe the direct report of the holdings and flows from every market participant. For example, the household sector in the Financial Accounts data is a residual sector imputed using market clearing.

After imposing market clearing, measurement errors in the observed data will negatively enter the quantity for the residual sector r:

$$\tilde{q}_{r,t} \equiv -\frac{1}{S_r}\sum_{i\neq r} S_i \tilde{q}_{i,t} = q_{r,t} - \frac{1}{S_r}\sum_{i\neq r} S_i \varepsilon_{i,t}.$$

In this case, moment conditions between the residual sector r and other sectors with measurement errors may no longer hold, as the measurement errors lead to a negative covariance:

$$\mathbb{E}\left[\left(u_{r,t} - \frac{1}{S_r}\sum_{i\neq r} S_i \varepsilon_{i,t}\right)\left(u_{i,t} + \varepsilon_{i,t}\right)\right] = -\frac{S_i}{S_r}\sigma^2_{i,\varepsilon}.$$

If one is not interested in the aggregate market elasticity, one can simply exclude the residual sector from the estimation. If there exist investors free from measurement errors $(\sigma_{i,\varepsilon} = 0)$, then the elasticity for the residual sector can also be consistently estimated by only including sectors without measurement errors.

S2.4.3 Uniqueness of the optimal GIV solution

For the optimal weighting matrix $\mathbf{W}_t^*$, the second part of Assumption A.2 follows from the first part in many cases. For example, when size weights and $\mathbf{C}_{i,t}$ are constant across time, or $\mathbf{C}_{i,t}$ are non-negative with probability 1 (such as dummy variables), non-singular conditions in $\mathbb{E}\left[\mathbf{W}_t^{*\prime}\mathbf{W}_t^*\right]$ automatically imply unique identification. The additional assumption is only needed when we have nonrestrictive loading factors $\mathbf{C}_{i,t}$. To see this, consider another candidate $\tilde{\boldsymbol{\zeta}} \equiv \boldsymbol{\zeta} + \Delta\boldsymbol{\zeta}$. Write $M_t := 1/(\mathbf{C}_{S,t}'\boldsymbol{\zeta})$ and let $\sigma_p^2 := \mathrm{Var}(p_t \mid$

$\mathbf{C}_t, \mathbf{S}_t$). We have:

$$
\begin{aligned}
\mathbb{E}\left[\mathbf{W}_t^{*\prime}\boldsymbol{\psi}_t\left(\tilde{\boldsymbol{\zeta}}\right)\right] &= \mathbf{0} \\
&= \mathbb{E}\left[M_t \sum_{i\neq j}\left(\frac{S_{jt}\mathbf{C}_{it}}{\sigma_i^2}+\frac{S_{it}\mathbf{C}_{jt}}{\sigma_j^2}\right)\left(u_{i,t}+p_t\mathbf{C}_{i,t}^{\prime}\Delta\boldsymbol{\zeta}\right)\left(u_{j,t}+p_t\mathbf{C}_{j,t}^{\prime}\Delta\boldsymbol{\zeta}\right)\right] \\
&= \mathbb{E}\left[M_t \sum_{i\neq j}\left(\frac{S_{jt}\mathbf{C}_{it}}{\sigma_i^2}+\frac{S_{it}\mathbf{C}_{jt}}{\sigma_j^2}\right)\left(M_t\left(S_{jt}\sigma_j^2\mathbf{C}_{i,t}^{\prime}+S_{it}\sigma_i^2\mathbf{C}_{j,t}^{\prime}\right)\Delta\boldsymbol{\zeta}+\sigma_p^2\Delta\boldsymbol{\zeta}^{\prime}\mathbf{C}_{i,t}\mathbf{C}_{j,t}^{\prime}\Delta\boldsymbol{\zeta}\right)\right] \\
&= \mathbb{E}\left[\mathbf{W}_t^{*\prime}\mathbf{V}^{\psi}\mathbf{W}_t^{*}\right]\Delta\boldsymbol{\zeta}+\mathbb{E}\left[\sigma_p^2 M_t \sum_{i\neq j}\left(\frac{S_{jt}\mathbf{C}_{it}}{\sigma_i^2}+\frac{S_{it}\mathbf{C}_{jt}}{\sigma_j^2}\right)\left(\Delta\boldsymbol{\zeta}^{\prime}\mathbf{C}_{i,t}\mathbf{C}_{j,t}^{\prime}\Delta\boldsymbol{\zeta}\right)\right].
\end{aligned}
$$

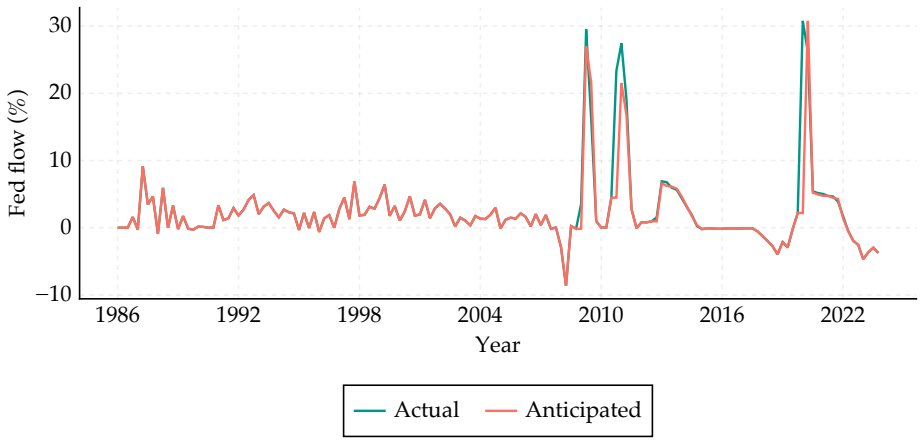
With non-negative $\mathbf{C}_{it}$ and non-singular $\mathbb{E}\left[\mathbf{W}_t^{*\prime}\mathbf{W}_t^{*}\right]$, the only solution to the equation above is $\Delta\boldsymbol{\zeta}=\mathbf{0}$. However, with unrestricted $\mathbf{C}_{i,t}$ potentially multiple roots can arise from this equation. One needs to verify the uniqueness of solution when complicated loading factors are used.

S3 Additional Empirical Exhibits

S3.1 Scheduled Federal Reserve purchases: exhibits

Appendix B.1.2 describes the construction of the scheduled purchases. The exhibits below illustrate it. Figure S3.1 plots the path of the Fed's flows together with the one-quarter-ahead anticipated path sequence. For most quarters, the realized purchases closely track the anticipated purchases; the major deviations occur on the entry and exit of the quantitative easing (QE) programs. Table S3.1 reports the anticipated purchases versus the actual purchases of the Fed during active QE periods.

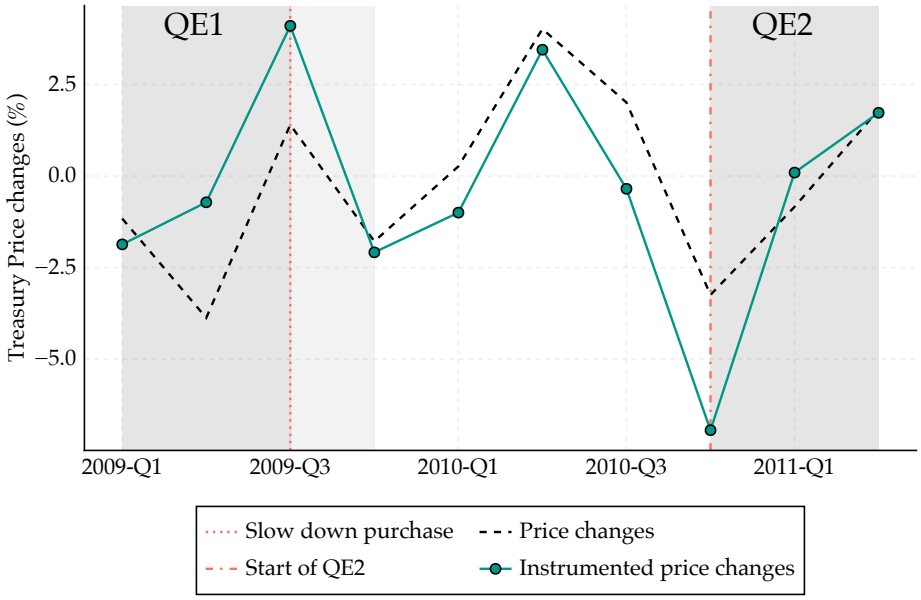
Figure S3.1: **Actual and anticipated Federal Reserve purchases**



Note: This figure plots actual Fed purchases of U.S. Treasury notes and bonds and the purchases anticipated one quarter ahead, normalized by total Fed holdings. Before 2009, anticipated purchases equal actual purchases; from the start of QE, they incorporate the New York Fed's pre-announced purchase schedule.

As a further illustration, in Figure S3.2 we focus on the end of the QE1 in 2009Q3 and the start of QE2 in 2010Q4. The original schedule for the first round of Treasury purchases ends at 2009Q3. In that quarter, the Federal Open Market Committee (FOMC) meetings decided to change the schedule and slow down the purchase and extend to Q4 in order to "promote a smooth transition in markets as these purchases of Treasury securities are completed". This results in a negative deviation from the anticipated path. In the figure, we plot the path of the price changes of the Treasury notes and bonds (the dash line), and the price change predicted using other sectors' demand shocks (the solid line). As shown in the price path, the quarter in which they decided to slow down the purchase is the one with the largest Treasury price appreciation in the entire QE1 episode. Similarly, the start of the QE2 also coincides with the quarter that saw the largest Treasury price depreciation.

Figure S3.2: **Case Study: QE Entry and Exit between 2009 and 2010**



Note: This figure plots observed and predicted quarterly Treasury price changes from 2009 to 2011, covering the end of QE1 and the start of QE2. The dashed line shows observed price changes. The solid line shows instrumented price changes, predicted using the size-weighted idiosyncratic shocks of non-Fed sectors extracted from the GMM estimation.

Table S3.1: The scheduled and actual purchases during QEs

	Anticipated ($ billions)	Actual ($ billions)	Remarks
2009-Q1		16	QE1 for Treasuries started (link)
2009-Q2	150	164	
2009-Q3	150	113	The FOMC decided to slow down the purchase (link)
2009-Q4	7	7	
2010-Q4		210	QE2 started (link)
2011-Q1	250	319	
2011-Q2	245	279	
2013-Q1	135	141	The FOMC announced in 2012-Q2 to QE3 (link)
2013-Q2	135	147	
2013-Q3	135	131	
2013-Q4	135	131	
2014-Q1	110	104	
2014-Q2	80	77	
2014-Q3	45	49	
2014-Q4	10	6	
2020-Q1		863	COVID-19 (link)
2020-Q2	863	1034	Total purchases not scheduled, use lagged purchases relative to holdings
2020-Q3	240	250	
2020-Q4	240	250	
2021-Q1	240	254	
2021-Q2	240	240	
2021-Q3	240	250	
2021-Q4	240	217	
2022-Q1	80	100	

S3.2 Case Study: Household and Mutual Fund Sectors

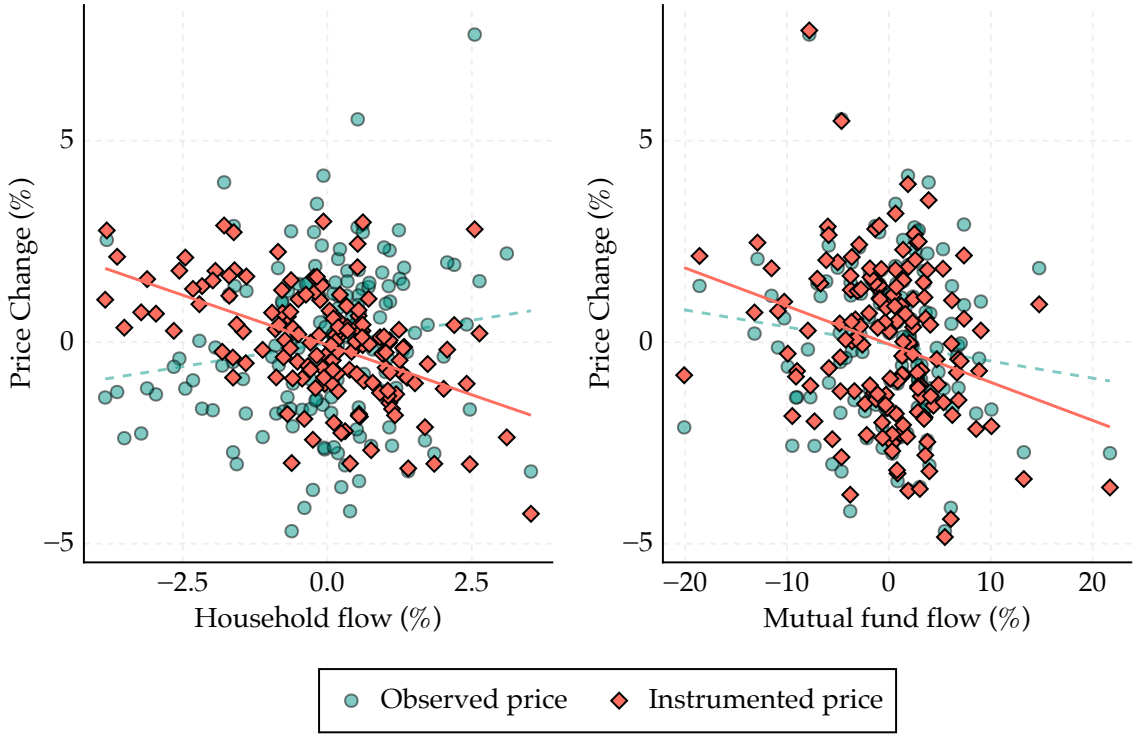
Figure S3.3 provides an illustration of our sector-specific estimates against the reduced-form OLS estimates. For the household sector and the mutual fund sector, we plot their "demand curves". We residualize sector inflow into Treasuries against common factors and plot it on the x-axis. The y-axis represents two notions of Treasury price changes. Using green markers, we plot the observed price changes, also residualized against the common factors. The observed-price relation slopes upward for households and slightly downward for mutual funds. Simultaneity, however, renders OLS estimators biased in this context: When sectors are granular and demand is downward-sloping, a higher asset price reduces demand, but the reduced demand will counteract the price.

In red, we plot the flows against the predicted price changes from idiosyncratic shocks from other sectors that we extract from our estimation. As discussed in the IV interpretation in Section 3.3, the red lines summarize the relation between residualized flows and instrumented prices over the sample.[7] Using the instrumented price changes reveals that the household sector is in fact highly elastic. Mutual funds are also slightly more elastic than what the simple correlation suggests.

The bias-corrected OLS interpretation of our optimal GIV estimator (equation (A.7) in Appendix A.1) helps interpret these differences. Though the mutual fund sector and the household sector are comparable in terms of sizes, the household sector has much more volatile idiosyncratic shocks than the mutual fund sector, a pattern that is also visible from the volatility of the raw flows. The optimal GIV estimator recognizes this and makes a larger correction for the household sector than for mutual funds.

[7] To further ensure the exclusion restriction is respected, we can exclude all moment conditions involving the target sectors (households/mutual funds) when extracting idiosyncratic shocks for other sectors. In this way, those extracted idiosyncratic shocks are completely unrelated to the target sectors except through the price. This corresponds to the leave-one-out estimator reported in Appendix Table S3.2.

Figure S3.3: **Optimal GIV vs. OLS: Two examples**



Note: This figure compares OLS and optimal GIV estimates of demand elasticities for households and mutual funds. In each panel, the horizontal axis shows the sector's net inflow into U.S. Treasury notes and bonds. The vertical axis shows quarterly Treasury price changes. Both flows and observed price changes are residualized against the common factors in the estimated specification. Green points use observed price changes; red points replace these with price changes predicted using size-weighted idiosyncratic shocks from the other sectors, extracted from the GMM estimation.

S3.3 Mapping Multiplier Estimates from the Literature

This appendix outlines how we map each paper's estimates to the multipliers plotted in Figure B.8.

Bernanke, Reinhart, and Sack (2004) examine how 10-year Treasury yields respond to Japanese Ministry of Finance interventions in the foreign exchange market between 2000 and 2004. Specifically, they study three-day windows around these interventions and find that a $1 billion purchase of dollars (subsequently invested in Treasuries) reduces the 10-year yield by 0.66 basis points. To express this result as a macro multiplier, we proceed in two steps. First, we scale their semi-elasticity to a 1% demand shock as a share of outstanding. The average amount of Treasury notes and bonds outstanding during the sample period was $2.4 trillion. Thus, a 1% demand shock corresponds to a $24 billion purchase. Since a $1 billion purchase lowers yields by 0.66 basis points, a 1% demand shock lowers yields by $0.66bp/100 \times 24 = 0.158$ percentage points. Second, we now map this semi-elasticity to a price multiplier. If we interpret the yield change as a level shift of the curve, we can approximate the price level impact on the Treasury market using the fact $\Delta P_t/P_t \approx -\text{Duration} \times \Delta y_t$. The average duration of the Treasury market is around 6.5 years. Thus, the implied macro multiplier is 1.03.

Warnock and Warnock (2009) They regress monthly 10-year Treasury yields on official inflows (as a share of lagged GDP) over the 1984–2005 period. Their estimates imply that a 1% flow shock (relative to GDP) lowers yields by 25 basis points. During this sample, outstanding Treasuries averaged 56% of GDP, so a 1% demand shock relative to outstanding supply corresponds to a decline in 10-year yields of $(25bp/100) \times 0.56 = 0.14$ percentage points. Assuming the yield change reflects a parallel shift of the yield curve, and using an average Treasury duration of 6.5 years, this implies a price impact multiplier of approximately 0.91.

Beltran, Kretchmer, Marquez, and Thomas (2013) use instruments (Japanese FX interventions, oil shocks, Chinese balance-of-payments surprises) to estimate the effect of official foreign purchases on the five-year term premium (1994–2007). A 1% demand shock (of outstanding) lowers the five-year term premium by 0.135 percentage points. Assuming this corresponds to a parallel shift in the yield curve and applying the duration approximation, this estimate maps to a macro multiplier of 0.88.

Wolcott (2020) estimate a VAR with macro variables, foreign flows, and yields, imposing no-arbitrage. Using Cholesky identification, they find a one-standard-deviation foreign flow shock (0.3% of outstanding) lowers the five-year yield by 2 basis points.

Scaling to a 1% shock gives a 0.067 percentage point yield effect. Using the duration approximation this implies a macro multiplier of 0.44.

Ahmed and Rebucci (2024) identify exogenous foreign official flow shocks via heteroskedasticity in a structural VAR. An unexpected \$100 billion purchase (1.03% of the \$9.7 trillion stock of Treasury notes and bonds held by the public at the end of the sample period[8]) lowers the 10-year yield by 125 basis points. Scaling to a 1% shock gives a 1.21 percentage point yield move. Applying the duration approximation, this implies a macro multiplier of 7.87.

Gagnon, Raskin, Remache, and Sack (2011) provide two estimates. First, an event study of QE1 announcements (2008–2009) finds that eight windows lowered the 10-year yield by 91 basis points. If the full \$300 billion (6.8% of the \$4.41 trillion stock) was a surprise revealed in these eight windows, this is a 0.134 percentage point yield effect per 1% shock. Using the duration approximation, the multiplier is 0.87. Second, a monthly regression of yields on supply (assumed exogenous) finds a 1% increase in supply-to-GDP raises the term premium by 4.4 basis points. With average debt/GDP of 57%, this is 0.025 percentage point per 1% shock (of outstanding). With the duration approximation, the multiplier is 0.16. The challenge with these supply shocks is that Treasury supply decisions are typically pre-announced so the price impact tends to happen before the actual supply change, resulting in understating the size of the multiplier.

Swanson (2011) uses the Kennedy administration's Operation Twist announcements (1961–1962) as exogenous shocks. The first four announcements lowered the 10-year yield by 16 basis points, with purchases equal to 4.6% of total marketable Treasury debt outstanding. Thus, a 1% shock moves yields by 0.035 percentage points. With the duration approximation this implies a macro multiplier of 0.23.

Kaminska and Zinna (2020) estimate the Vayanos–Vila preferred-habitat model via Bayesian MCMC and simulate the three LSAP waves. The implied multipliers are $M_1 = 0.76$, $M_2 = 1.32$, and $M_3 = 0.45$, with an average of 0.84.

Jansen, Li, and Schmid (2024) estimate investor demand (pension funds, mutual funds, insurance companies, etc.) for short-, medium-, and long-term Treasuries, and embed these in a Vayanos and Vila (2021)-style model, treating broker-dealers and hedge funds as arbitrageurs. They estimate a macro multiplier of 1.25 for a permanent 1% demand shock (focusing on medium- and long-term Treasuries for comparability). We focus

[8] Financial Accounts table L.210, 2017Q3: total marketable notes-and-bonds liability less Federal Reserve SOMA holdings.

on their permanent demand shock, as our analysis also considers permanent changes in holdings.

A few notable papers examine the effects of exogenous quantity changes in the Treasury market, though their results are not directly comparable to ours.

For example, Krishnamurthy and Vissing-Jorgensen (2012) study how Treasury supply affects the convenience yield. They find that, between 1969 and 2008, a 1% increase in the debt-to-GDP ratio reduces the convenience yield by 0.87 basis points (measured using 10-year yields). Adjusting for the average debt-to-GDP ratio during this period and using the duration approximation, this implies a multiplier of 0.02-0.03. This smaller multiplier reflects two key differences: their dependent variable is the convenience yield (not the level of yields), and their analysis is conducted at an annual frequency (rather than quarterly). When we estimate the impact of our demand shocks on the convenience yield at an annual frequency, we obtain a similar estimate.

Similarly, Greenwood, Hanson, and Stein (2015) show that short-term debt yields rise when Treasury bill supply increases, consistent with a downward-sloping demand curve for T-bills. However, their object of interest differs from ours: they focus on the effect of Treasury bill supply on short-term debt yields, while we study the effect of demand on Treasury Notes and Bonds yields (long-term debt).

S3.4 Leave-one-out tables

Tables S3.2 and S3.3 report the estimates plotted in Figures B.2 and B.3.

Table S3.2: **Price elasticity by sector: Leave-one-out Estimation**

Left out	Households†	Rest of world	Banks	Dealers†	Fed	Pension funds	Other	Mutual funds	Insurance	ETFs	Supply†	ζ_S
None	0.35***	0.35***	1.46***	0.06**	1.21***	0.41***	0.21**	0.59***	0.44***	−0.20	0.11***	1.01***
	(0.06)	(0.07)	(0.12)	(0.03)	(0.11)	(0.09)	(0.09)	(0.16)	(0.09)	(0.17)	(0.03)	(0.06)
Households	–	0.51***	1.22***	0.02	1.01	0.41**	0.20	0.54	0.47**	−0.16	0.10	0.98***
		(0.16)	(0.36)	(0.09)	(0.67)	(0.18)	(0.15)	(0.37)	(0.20)	(0.39)	(0.08)	(0.09)
Rest of world	0.39**	–	1.36***	0.04	1.22**	0.49**	0.05	0.67**	0.41*	−0.53	0.08**	0.97***
	(0.16)		(0.23)	(0.09)	(0.54)	(0.24)	(0.17)	(0.28)	(0.22)	(0.39)	(0.04)	(0.12)
Banks	0.39**	0.33***	–	0.07	0.94*	0.44**	0.17	0.69***	0.61**	−0.08	0.08*	1.03***
	(0.19)	(0.11)		(0.10)	(0.50)	(0.20)	(0.19)	(0.27)	(0.25)	(0.60)	(0.05)	(0.13)
Dealers	0.24	0.33**	1.50***	–	1.29**	0.48	0.20	0.47*	0.31	0.03	0.22	1.01***
	(0.19)	(0.13)	(0.28)		(0.61)	(0.30)	(0.15)	(0.24)	(0.21)	(0.44)	(0.26)	(0.09)
Fed	0.33**	0.36***	1.32***	0.07	–	0.46***	0.26*	0.45*	0.38**	−0.11	0.14***	1.02***
	(0.13)	(0.11)	(0.24)	(0.07)		(0.18)	(0.14)	(0.25)	(0.19)	(0.35)	(0.05)	(0.10)
Pension funds	0.34***	0.34***	1.45***	0.07	1.15**	–	0.19	0.61**	0.45**	−0.18	0.11**	1.01***
	(0.12)	(0.11)	(0.24)	(0.08)	(0.53)		(0.14)	(0.25)	(0.21)	(0.36)	(0.04)	(0.09)
Other	0.40***	0.25**	1.57***	0.02	1.15**	0.23	–	0.65**	0.50***	−0.33	0.12**	0.99***
	(0.14)	(0.11)	(0.25)	(0.06)	(0.54)	(0.16)		(0.27)	(0.17)	(0.31)	(0.05)	(0.12)
Mutual funds	0.38**	0.36***	1.51***	0.05	0.94**	0.39**	0.21	–	0.59***	0.01	0.10**	1.02***
	(0.16)	(0.10)	(0.24)	(0.07)	(0.47)	(0.18)	(0.15)		(0.22)	(0.56)	(0.05)	(0.12)
Insurance	0.36***	0.33***	1.52***	0.05	1.13**	0.42**	0.22	0.66***	–	−0.19	0.10**	1.00***
	(0.13)	(0.10)	(0.24)	(0.07)	(0.52)	(0.18)	(0.15)	(0.25)		(0.36)	(0.04)	(0.10)
ETFs	0.35***	0.35***	1.45***	0.06	1.15**	0.42**	0.20	0.60**	0.44**	–	0.11**	1.00***
	(0.13)	(0.10)	(0.23)	(0.07)	(0.53)	(0.18)	(0.14)	(0.25)	(0.19)		(0.04)	(0.10)
Supply	0.21	0.40***	1.23***	0.19	1.77*	0.31	0.36**	0.60**	0.20	−0.86***	–	1.04***
	(0.24)	(0.13)	(0.24)	(0.45)	(0.91)	(0.24)	(0.15)	(0.28)	(0.18)	(0.27)		(0.18)

Note: This table reports leave-one-out estimates of the price elasticities in Table 1. Each row re-estimates the specification without the moment conditions of the named sector; each column reports a sector's price elasticity ζ_i, averaged over its observed quarters. The columns for the Fed and ETFs cover 2009Q1–2023Q4. Sectors marked † have unit size weights, so their coefficients measure demand responses relative to the total market; the other sectors' coefficients are relative to their own holdings. The first row uses all sectors and standard errors from equation (A.5); the other rows use the sandwich variance in equation (A.6). Standard errors are in parentheses below estimates. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table S3.3: **Change in price elasticities across 2009Q1: Leave-one-out Estimation**

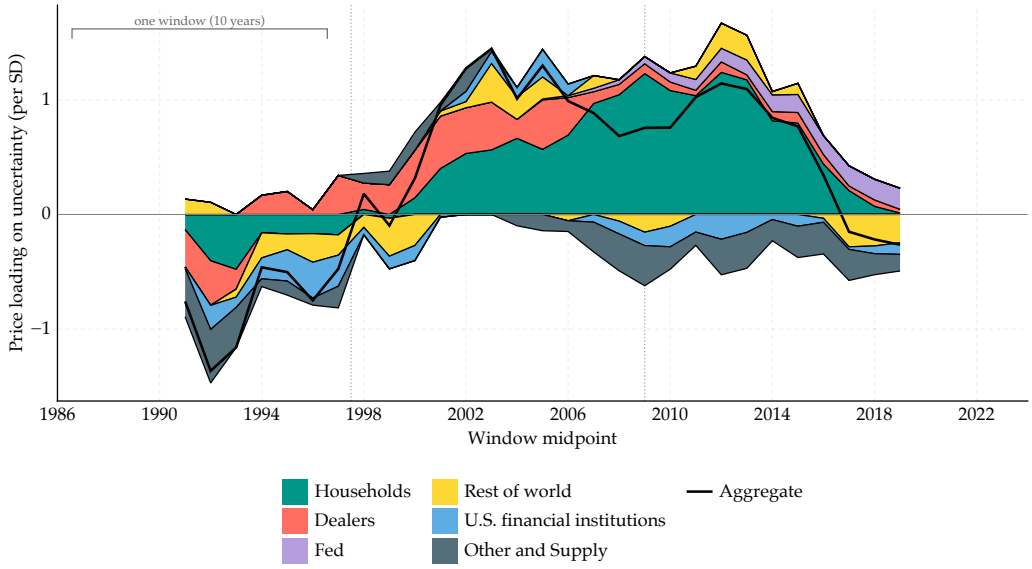
	Elasticity change, post−pre 2009Q1										Aggregate ζ_S		
Left out	Households†	Rest of world	Banks	Dealers†	Fed	Pension funds	Other	Mutual funds	Insurance	Supply†	Pre	Post	Change
None	−0.16	−0.43***	−0.57**	−0.06	1.21***	−0.33**	0.55***	−0.25	−0.06	−0.06	1.14***	0.82***	−0.32***
	(0.11)	(0.13)	(0.23)	(0.06)	(0.11)	(0.16)	(0.17)	(0.29)	(0.16)	(0.05)	(0.10)	(0.07)	(0.12)
Households	–	−0.73**	−1.04	−0.01	1.01	−0.37	0.57*	0.73	−0.03	−0.09	1.14***	0.74***	−0.40**
		(0.30)	(0.80)	(0.15)	(0.67)	(0.33)	(0.29)	(0.76)	(0.37)	(0.14)	(0.11)	(0.13)	(0.17)
Rest of world	−0.29	–	−0.44	−0.06	1.22**	−0.12	0.50	−0.69	−0.06	−0.02	1.13***	0.73***	−0.40*
	(0.28)		(0.48)	(0.16)	(0.54)	(0.43)	(0.32)	(0.51)	(0.40)	(0.08)	(0.18)	(0.11)	(0.21)
Banks	−0.09	−0.38*	–	−0.09	0.94*	−0.30	0.55	0.16	0.02	−0.05	1.14***	0.86***	−0.28
	(0.41)	(0.21)		(0.16)	(0.50)	(0.38)	(0.38)	(0.54)	(0.52)	(0.09)	(0.15)	(0.23)	(0.28)
Dealers	−0.04	−0.42*	−0.73	–	1.29**	−0.44	0.54*	−0.12	0.11	−0.24	1.16***	0.79***	−0.36**
	(0.34)	(0.24)	(0.58)		(0.61)	(0.51)	(0.29)	(0.46)	(0.38)	(0.44)	(0.14)	(0.11)	(0.17)
Fed	−0.23	−0.41*	−0.90*	−0.03	–	−0.24	0.67**	−0.62	−0.21	0.03	1.13***	0.84***	−0.29
	(0.24)	(0.21)	(0.51)	(0.12)		(0.32)	(0.28)	(0.49)	(0.34)	(0.12)	(0.15)	(0.11)	(0.19)
Pension funds	−0.14	−0.41**	−0.58	−0.08	1.15**	–	0.52*	−0.22	−0.01	−0.08	1.14***	0.80***	−0.34*
	(0.23)	(0.21)	(0.51)	(0.13)	(0.53)		(0.28)	(0.47)	(0.38)	(0.08)	(0.13)	(0.13)	(0.18)
Other	−0.26	−0.31	−0.75	0.01	1.15**	−0.37	–	−0.20	−0.19	−0.02	1.10***	0.82***	−0.28
	(0.27)	(0.20)	(0.55)	(0.10)	(0.54)	(0.30)		(0.51)	(0.32)	(0.10)	(0.18)	(0.14)	(0.23)
Mutual funds	0.01	−0.47**	−0.37	−0.06	0.94**	−0.39	0.41	–	0.18	−0.08	1.12***	0.87***	−0.25
	(0.34)	(0.21)	(0.53)	(0.12)	(0.47)	(0.35)	(0.33)		(0.46)	(0.09)	(0.15)	(0.20)	(0.25)
Insurance	−0.18	−0.38*	−0.69	−0.05	1.13**	−0.37	0.47	−0.29	–	−0.04	1.14***	0.80***	−0.34*
	(0.25)	(0.20)	(0.51)	(0.12)	(0.52)	(0.33)	(0.29)	(0.47)		(0.08)	(0.15)	(0.13)	(0.20)
ETFs	−0.18	−0.42**	−0.59	−0.06	1.15**	−0.36	0.53*	−0.23	−0.05	−0.05	1.13***	0.80***	−0.33*
	(0.24)	(0.20)	(0.50)	(0.12)	(0.53)	(0.32)	(0.28)	(0.47)	(0.35)	(0.08)	(0.15)	(0.13)	(0.19)
Supply	−0.20	−0.37	−0.24	−0.29	1.77*	−0.32	0.58**	−0.52	−0.19	–	1.19***	0.80***	−0.39
	(0.41)	(0.25)	(0.55)	(0.74)	(0.91)	(0.42)	(0.28)	(0.52)	(0.32)		(0.28)	(0.13)	(0.31)

Note: This table reports the post- minus pre-2009 change in the sector price elasticities ζ_i from the leave-one-out estimates of Table S3.2, and the aggregate elasticity ζ_S before and after 2009Q1. Sectors marked † have unit size weights, so their coefficients measure demand responses relative to the total market; the other sectors' coefficients are relative to their own holdings. Standard errors are in parentheses below estimates, computed as in Table S3.2. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

S3.5 Safe haven: direct contributions and sector detail

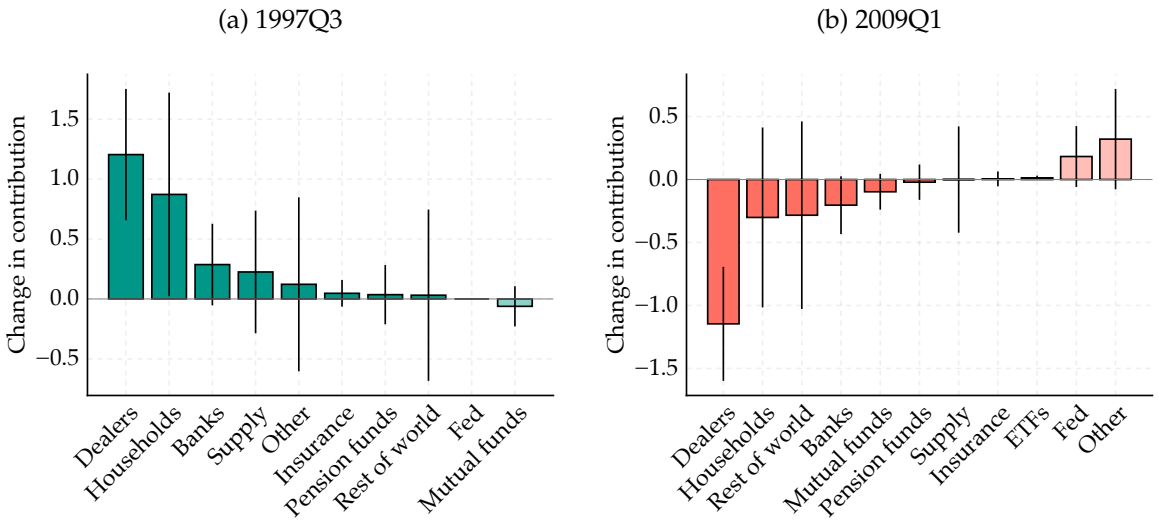
Figures S3.4 and S3.5 are the direct-loading counterparts discussed in Appendix B.3.3. Table S3.4 reports each sector's price elasticity and its direct and total uncertainty responses in the first (panel a) and third (panel b) regimes.

Figure S3.4: **Ten-year rolling contributions to the safe haven: direct loadings**



Note: This figure plots investor groups' contributions to the price loading on uncertainty using direct loadings; see the note to Figure 4 for details on the rolling estimation and sector definitions. Windows are plotted at their midpoints; dotted vertical lines mark the regime shifts at 1997Q3 and 2009Q1.

Figure S3.5: **Changes in direct uncertainty contributions**



Note: This figure plots changes in each sector's contribution to the aggregate direct uncertainty loading at 1997Q3 (panel (a)) and 2009Q1 (panel (b)), using the specification of Figure 5. Sectors, including supply, are shown separately and ranked by their changes. Whiskers are 95% confidence intervals, from equation (A.9).

Table S3.4: **Sector elasticities and uncertainty responses in the first and third regimes**

(a) 1986Q1–1997Q2

	Elasticity		Direct uncertainty		Total uncertainty	
	Loading	Contribution	Loading	Contribution	Loading	Contribution
Aggregate	1.04*** [0.76, 1.33]	—	−1.12** [−2.20, −0.03]	—	−0.35 [−1.77, 1.07]	—
Households†	0.33** [0.03, 0.64]	0.33** [0.03, 0.64]	−0.15 [−0.77, 0.46]	−0.15 [−0.77, 0.46]	−0.02 [−0.63, 0.58]	−0.02 [−0.63, 0.58]
Dealers†	−0.11 [−0.24, 0.03]	−0.11 [−0.24, 0.03]	−0.13 [−0.52, 0.26]	−0.13 [−0.52, 0.26]	−0.10 [−0.50, 0.31]	−0.10 [−0.50, 0.31]
Rest of world	0.64*** [0.35, 0.93]	0.13*** [0.07, 0.19]	−0.16 [−1.56, 1.24]	−0.03 [−0.32, 0.25]	0.28 [−1.09, 1.64]	0.06 [−0.22, 0.34]
Fed						
Banks	1.81*** [1.36, 2.27]	0.28*** [0.21, 0.35]	−0.84 [−2.61, 0.93]	−0.13 [−0.40, 0.14]	−0.09 [−2.32, 2.14]	−0.01 [−0.35, 0.33]
Pension funds	1.00*** [0.60, 1.40]	0.18*** [0.11, 0.25]	−0.40 [−1.64, 0.83]	−0.07 [−0.29, 0.15]	0.48 [−0.87, 1.84]	0.08 [−0.15, 0.32]
Insurance	0.21 [−0.18, 0.60]	0.02 [−0.02, 0.06]	−0.69 [−1.74, 0.36]	−0.06 [−0.16, 0.03]	−0.06 [−1.18, 1.06]	−0.01 [−0.11, 0.10]
Mutual funds	0.70** [0.02, 1.37]	0.04** [0.00, 0.08]	−0.17 [−2.39, 2.04]	−0.01 [−0.14, 0.12]	0.03 [−2.08, 2.15]	0.00 [−0.12, 0.12]
Other	−0.01 [−0.42, 0.41]	0.00 [−0.09, 0.08]	−1.62 [−4.38, 1.15]	−0.33 [−0.91, 0.24]	−1.00 [−3.45, 1.44]	−0.21 [−0.71, 0.30]
Supply†	0.17*** [0.05, 0.30]	0.17*** [0.05, 0.30]	−0.19 [−0.55, 0.18]	−0.19 [−0.55, 0.18]	−0.15 [−0.48, 0.18]	−0.15 [−0.48, 0.18]

(b) 2009Q1–2023Q4

	Elasticity		Direct uncertainty		Total uncertainty	
	Loading	Contribution	Loading	Contribution	Loading	Contribution
Aggregate	0.83*** [0.70, 0.97]	—	0.11 [−0.65, 0.88]	—	0.43 [−0.70, 1.55]	—
Households†	0.15*** [0.04, 0.26]	0.15*** [0.04, 0.26]	0.42** [0.05, 0.78]	0.42** [0.05, 0.78]	0.40** [0.06, 0.74]	0.40** [0.06, 0.74]
Dealers†	0.04 [−0.02, 0.11]	0.04 [−0.02, 0.11]	−0.07 [−0.31, 0.17]	−0.07 [−0.31, 0.17]	0.03 [−0.22, 0.29]	0.03 [−0.22, 0.29]
Rest of world	0.15* [0.00, 0.31]	0.07* [0.00, 0.14]	−0.61 [−1.66, 0.43]	−0.29 [−0.77, 0.20]	−0.80 [−1.77, 0.17]	−0.37 [−0.82, 0.08]
Fed	1.18*** [0.96, 1.40]	0.26*** [0.21, 0.31]	0.83 [−0.27, 1.94]	0.18 [−0.06, 0.42]	2.18 [−0.61, 4.97]	0.48 [−0.13, 1.09]
Banks	1.06*** [0.79, 1.32]	0.05*** [0.04, 0.07]	−0.91 [−3.07, 1.26]	−0.05 [−0.15, 0.06]	−0.10 [−2.05, 1.86]	0.00 [−0.10, 0.09]
Pension funds	0.17* [−0.01, 0.36]	0.01* [0.00, 0.02]	−1.07*** [−1.81, −0.34]	−0.06*** [−0.10, −0.02]	−1.35*** [−2.05, −0.65]	−0.07*** [−0.11, −0.03]
Insurance	0.28*** [0.09, 0.46]	0.01*** [0.00, 0.01]	−0.47 [−1.14, 0.19]	−0.01 [−0.03, 0.01]	−0.50 [−1.15, 0.16]	−0.01 [−0.03, 0.00]
Mutual funds	0.34* [0.00, 0.68]	0.02* [0.00, 0.05]	−2.51*** [−3.98, −1.05]	−0.17*** [−0.27, −0.07]	−1.98*** [−3.38, −0.59]	−0.13*** [−0.23, −0.04]
ETFs	−0.34** [−0.67, −0.01]	0.00** [−0.01, 0.00]	1.19 [−0.33, 2.70]	0.01 [0.00, 0.03]	1.36** [0.11, 2.60]	0.02** [0.00, 0.03]
Other	0.68*** [0.49, 0.87]	0.08*** [0.06, 0.10]	0.94 [−0.51, 2.39]	0.11 [−0.06, 0.28]	0.60 [−0.71, 1.92]	0.07 [−0.08, 0.22]
Supply†	0.15*** [0.09, 0.21]	0.15*** [0.09, 0.21]	0.04 [−0.19, 0.26]	0.04 [−0.19, 0.26]	0.02 [−0.19, 0.23]	0.02 [−0.19, 0.23]

Note: This table reports each sector's price elasticity ζ_i and its direct and total uncertainty loadings λ_i^{unc}, each with its contribution, $S_i\zeta_i$ or $S_i\lambda_i^{unc}$, from the specification of Table 3; each panel covers one regime. The Banks, Pension funds, Insurance, Mutual funds, ETFs, and Other rows make up Remaining domestic in Table 3; ETFs enter the sample in 2009Q1. Sectors marked † have unit size weights, so their coefficients measure demand responses relative to the total market; the other sectors' coefficients are relative to their own holdings. The Federal Reserve's pre-2009 purchases are treated as fully anticipated through the scheduled-purchase control, so its pre-2009 coefficients are not estimated and are left blank. 95% confidence intervals, computed as in Tables 1 and 3, are in brackets below estimates. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

S4 Proofs

This appendix proves the results stated in Appendix A and Theorem 1 of the main text.

S4.1 Proof of Theorem A.1

In this subsection, we proceed as if there were no exogenous variables $\mathbf{X}_{i,t}$. As we show in Appendix A.2, all results described here go through with observed common factors.

We proceed in the following steps. We first derive the asymptotic properties for a broad class of GMM estimators with generic weighting matrices. Then we derive the semiparametric efficiency bound for the class of GMM estimators. The first two steps follow the standard GMM literature. Finally, we show that the optimal GIV estimator in Equation (A.4) achieves the semiparametric efficiency bound.

S4.1.1 Asymptotic Properties of GMM Estimators

For ease of reading, we restate the moment conditions of the model here.

Let $\mathbf{z}$ be a generic parameter vector. The full moment conditions of the model can be summarized with a vector $\boldsymbol{\psi}_t(\mathbf{z})$ of length $\frac{N(N-1)}{2}$ stacking pairs with indices $i < j$ (we take $i < j$ to avoid duplicating symmetric cross-moments); the subscript ij corresponds to the entity pair (i, j):

$$\psi_{ij,t}(\mathbf{z}) \equiv \hat{u}_{i,t}(\mathbf{z})\,\hat{u}_{j,t}(\mathbf{z}) \equiv \left(q_{i,t} + p_t \mathbf{C}'_{i,t}\mathbf{z}\right)\left(q_{j,t} + p_t \mathbf{C}'_{j,t}\mathbf{z}\right).$$

We have the following moment conditions that under the true parameter $\boldsymbol{\zeta}$, the conditional expectation of $\boldsymbol{\psi}$ is zero:

$$\begin{aligned}\mathbb{E}\left[\psi_{ij,t}(\boldsymbol{\zeta}) \mid \mathbf{C}_t, \mathbf{S}_t\right] &= \mathbb{E}\left[\left(q_{i,t} + p_t \mathbf{C}'_{i,t}\boldsymbol{\zeta}\right)\left(q_{j,t} + p_t \mathbf{C}'_{j,t}\boldsymbol{\zeta}\right) \mid \mathbf{C}_t, \mathbf{S}_t\right] \\ &= \mathbb{E}\left[u_{i,t}u_{j,t} \mid \mathbf{C}_t, \mathbf{S}_t\right] = \mathbf{0}. \end{aligned} \tag{S4.1}$$

Notice that this is an overidentified system, as we have $\frac{N(N-1)}{2}$ moment conditions while N_ζ parameters to be estimated.

Consider the class of GMM estimators $\hat{\boldsymbol{\zeta}}^W$ indexed by the weighting matrix $\mathbf{W}_t(\mathbf{z})$. The weighting matrix is measurable with respect to the filtration generated by $\mathbf{C}_t$ and $\mathbf{S}_t$. The estimator $\hat{\boldsymbol{\zeta}}^W$ solves:[9]

[9]This formulation nests the standard $\hat{\boldsymbol{\zeta}}^A = \arg\min_{\boldsymbol{z}} \sum_t \boldsymbol{\psi}_t(\boldsymbol{z})'\mathbf{A}\boldsymbol{\psi}_t(\boldsymbol{z})$, which is equivalent to a time-

$$\mathbf{b}_T^{\mathbf{W}}\left(\hat{\boldsymbol{\zeta}}^W\right) \equiv \frac{1}{T}\sum_t W_t\left(\hat{\boldsymbol{\zeta}}^W\right)'\boldsymbol{\psi}_t\left(\hat{\boldsymbol{\zeta}}^W\right) = 0. \tag{S4.2}$$

As a shorthand, we also define $\mathbf{W}_t \equiv \mathbf{W}_t(\boldsymbol{\zeta})$ as the weighting scheme evaluated at the true parameter.

Assumption A.2 ensures the estimator is well-behaved.

The lemma below characterizes the asymptotic properties of the GMM estimator $\hat{\boldsymbol{\zeta}}^W$.

Lemma S4.1 (Asymptotic Properties of GMM Estimator.)**.** *Under Assumption A.2, the GMM estimator $\hat{\boldsymbol{\zeta}}^W$ is consistent and asymptotically normal, and its asymptotic variance is given by:*

$$\mathbf{V}_\zeta^W = \mathbb{E}\left[\mathbf{W}_t'\mathbf{D}_t\right]^{-1}\mathbb{E}\left[\mathbf{W}_t'\mathbf{V}^\psi\mathbf{W}_t\right]\mathbb{E}\left[\mathbf{D}_t'\mathbf{W}_t\right]^{-1} \tag{S4.3}$$

where $\mathbf{J}_t = \frac{\partial\boldsymbol{\psi}_t(\mathbf{z})}{\partial\mathbf{z}}\,|_{\mathbf{z}=\boldsymbol{\zeta}}$ is the (raw) Jacobian matrix of $\boldsymbol{\psi}_t(\mathbf{z})$ evaluated at the true parameter, we define $\mathbf{D}_t \equiv \mathbb{E}[\mathbf{J}_t \mid \mathbf{C}_t, \mathbf{S}_t]$, and $\mathbf{V}^\psi \equiv \mathbb{E}\left[\boldsymbol{\psi}_t(\boldsymbol{\zeta})\boldsymbol{\psi}_t(\boldsymbol{\zeta})'\right] = \operatorname{diag}\left(\sigma_1^2\sigma_2^2, \sigma_1^2\sigma_3^2, \ldots, \sigma_{N-1}^2\sigma_N^2\right)$ is the covariance matrix of $\boldsymbol{\psi}_t(\boldsymbol{\zeta})$.

Proof. The consistency of the estimator directly follows the standard GMM argument. To derive the asymptotic variance, following the standard GMM formula, we have:

$$\mathbf{V}_\zeta^W = \mathbb{E}\left[\mathbf{D}_{W,t}\right]^{-1}\mathbb{E}\left[\mathbf{W}_t'\boldsymbol{\psi}_t(\boldsymbol{\zeta})\boldsymbol{\psi}_t(\boldsymbol{\zeta})'\mathbf{W}_t\right]\mathbb{E}\left[\mathbf{D}_{W,t}'\right]^{-1}$$

where $\mathbf{D}_{W,t} = \frac{\partial(\mathbf{W}_t(\mathbf{z})'\boldsymbol{\psi}_t(\mathbf{z}))}{\partial\mathbf{z}}\,|_{z=\zeta}$. Evaluating $\mathbf{D}_{W,t}$ at $\mathbf{z}=\boldsymbol{\zeta}$, we have:

$$\mathbb{E}\left[\mathbf{D}_{W,t}\right] = \mathbb{E}\left[\frac{\partial\mathbf{W}_t(\mathbf{z})'}{\partial\mathbf{z}}\boldsymbol{\psi}_t(\boldsymbol{\zeta}) + \mathbf{W}_t'\frac{\partial\boldsymbol{\psi}_t(\mathbf{z})}{\partial\mathbf{z}}\right]\Bigg|_{\mathbf{z}=\boldsymbol{\zeta}} = \mathbb{E}\left[\mathbf{W}_t'\mathbf{D}_t\right]$$

Using the law of iterated expectations, we have: $\mathbb{E}\left[\mathbf{W}_t'\boldsymbol{\psi}_t(\boldsymbol{\zeta})\boldsymbol{\psi}_t(\boldsymbol{\zeta})'\mathbf{W}_t\right] = \mathbb{E}\left[\mathbf{W}_t'\mathbf{V}^\psi\mathbf{W}_t\right]$. Plugging them in, we obtain Equation (S4.3). □

S4.1.2 Semiparametric Efficiency Bound

The semi-parametric efficiency bound for the class of GMM estimators is given by the following lemma:

Lemma S4.2 (Semiparametric efficiency bound)**.** *Under Assumptions A.1-A.2, the asymptotic variance of the GMM estimator $\hat{\boldsymbol{\zeta}}^W$ is minimized at the optimal weighting matrix $\mathbf{W}_t^*$ given*

invariant weighting matrix for the moment conditions after taking the first order condition. As the moment conditions are conditional expectations, the efficient weighting matrix is typically time-varying. See Newey (1993).

by:

$$\mathbf{W}_t^* = \left(\mathbf{V}^\psi\right)^{-1} \mathbf{D}_t, \tag{S4.4}$$

and the lower bound of the asymptotic variance is given by:

$$\mathbf{V}_\zeta^{W*} = \mathbb{E}\left[\mathbf{D}_t' \left(\mathbf{V}^\psi\right)^{-1} \mathbf{D}_t\right]^{-1} = \mathbb{E}\left[\mathbf{W}_t^{*\prime} \mathbf{V}^\psi \mathbf{W}_t^*\right]^{-1}. \tag{S4.5}$$

Proof. The proof for the optimality generally follows the strategy of optimal instrument in conditional models, surveyed in Newey (1993).

To show Equation (S4.5), note that since $\mathbf{W}_t^* = (\mathbf{V}^\psi)^{-1}\mathbf{D}_t$ and $\mathbf{V}^\psi$ is nonrandom (diagonal in $\sigma_i^2\sigma_j^2$), we have the pointwise identity

$$\mathbf{D}_t'(\mathbf{V}^\psi)^{-1}\mathbf{D}_t = \mathbf{W}_t^{*\prime}\mathbf{V}^\psi\mathbf{W}_t^*.$$

Taking expectations on both sides yields Equation (S4.5).

To show $\mathbf{V}_\zeta^{W*}$ is the lower bound of the asymptotic variance, we need to show that for any $\mathbf{W}_t$, $\mathbf{V}_\zeta^W - \mathbf{V}_\zeta^{W*}$ is positive semi-definite. Given the covariance matrices are symmetric positive definite, it is equivalent to show $\mathbf{\Delta} \equiv \left(\mathbf{V}_\zeta^{W*}\right)^{-1} - \left(\mathbf{V}_\zeta^W\right)^{-1}$ is positive semi-definite:

$$\mathbf{\Delta} \equiv \mathbb{E}\left[\mathbf{D}_t' \left(\mathbf{V}^\psi\right)^{-1} \mathbf{D}_t\right] - \mathbb{E}\left[\mathbf{D}_t'\mathbf{W}_t\right] \mathbb{E}\left[\mathbf{W}_t'\mathbf{V}^\psi\mathbf{W}_t\right]^{-1} \mathbb{E}\left[\mathbf{W}_t'\mathbf{D}_t\right].$$

Given $\mathbf{V}^\psi$ is a positive-definite diagonal matrix, it is convenient to factor $\mathbf{V}^\psi$ into $\mathbf{D}_t$ and $\mathbf{W_t}$ by defining:

$$\tilde{\mathbf{D}}_t = \left(\mathbf{V}^\psi\right)^{-\frac{1}{2}} \mathbf{D}_t$$
$$\tilde{\mathbf{W}}_t = \left(\mathbf{V}^\psi\right)^{\frac{1}{2}} \mathbf{W}_t.$$

Define $\mathbf{G}_t = \begin{bmatrix} \tilde{\mathbf{D}}_t' \\ \tilde{\mathbf{W}}_t' \end{bmatrix}$, and $\mathbf{H}' = \begin{bmatrix} \mathbf{I} & -\mathbb{E}\left[\tilde{\mathbf{D}}_t'\tilde{\mathbf{W}}_t\right] \mathbb{E}\left[\tilde{\mathbf{W}}_t'\tilde{\mathbf{W}}_t\right]^{-1} \end{bmatrix}$, we can express $\mathbf{\Delta}$ as:

$$\mathbf{\Delta} = \mathbf{H}'\mathbb{E}\left[\mathbf{G}_t\mathbf{G}_t'\right]\mathbf{H}.$$

Clearly, $\mathbf{\Delta}$ is positive semi-definite. Moreover, $\mathbf{\Delta} = 0$ when $\tilde{\mathbf{W}}_t = \tilde{\mathbf{D}}_t$, or $\mathbf{W}_t = \left(\mathbf{V}^\psi\right)^{-1} \mathbf{D}_t$. □

S4.1.3 Deriving the optimal GIV estimator

The remaining step is to show Equation (S4.4) gives the expression used in the optimal GIV estimator in Equation (A.3).

To show that, let $\mathbf{J}_t \equiv \frac{\partial \boldsymbol{\psi}_t(\mathbf{z})}{\partial \mathbf{z}} \mid_{\mathbf{z}=\boldsymbol{\zeta}}$ denote the raw Jacobian evaluated at the true parameter. Its entry is given as:

$$J_{ij,k,t} = u_{jt} p_t C_{it,k} + u_{it} p_t C_{jt,k},$$

Note that from the model structure, we have:

$$p_t = \frac{1}{\mathbf{C}'_{S,t}\boldsymbol{\zeta}} \left[\mathbf{X}'_{S,t}\boldsymbol{\beta} + u_{S,t}\right]$$

Hence we have:

$$\mathbb{E}\left[u_{j,t} p_t \mid \mathbf{C}_t, \mathbf{S}_t\right] = \frac{1}{\mathbf{C}'_{S,t}\boldsymbol{\zeta}} S_{j,t}\sigma_j^2.$$

Therefore, since $\mathbf{D}_t = \mathbb{E}[\mathbf{J}_t \mid \mathbf{C}_t, \mathbf{S}_t]$, we have

$$D_{ij,k,t} = \mathbb{E}\left[J_{ij,k,t} \mid \mathbf{C}_t, \mathbf{S}_t\right] = \frac{1}{\mathbf{C}'_{S,t}\boldsymbol{\zeta}} \left[S_{j,t}\sigma_j^2 C_{it,k} + S_{i,t}\sigma_i^2 C_{jt,k}\right].$$

Recall that $\mathbf{V}^{\psi} \equiv \mathbb{E}\left[\boldsymbol{\psi}_t(\boldsymbol{\zeta})\,\boldsymbol{\psi}_t(\boldsymbol{\zeta})'\right] = \operatorname{diag}\left(\sigma_1^2\sigma_2^2, \sigma_1^2\sigma_3^2, \ldots, \sigma_{N-1}^2\sigma_N^2\right)$. Plug into Equation (S4.4), we have:

$$\mathbf{W}^*_{ij,k,t} = \left(V_{ij}^{\psi}\right)^{-1} D_{ij,k,t} = \frac{1}{\mathbf{C}'_{S,t}\boldsymbol{\zeta}} \left(\frac{S_{j,t}C_{it,k}}{\sigma_i^2} + \frac{S_{i,t}C_{jt,k}}{\sigma_j^2}\right), \tag{S4.6}$$

which is exactly the weighting matrix used in the optimal GIV estimator in Equation (A.3).

Proof of Theorem 1. Finally, the proof to the Theorem 1 in the main text is a special case of the proof above, where $S_{jt} = S_j$ and $C_{it,k} = \begin{cases} 1 & i = k \\ 0 & i \neq k \end{cases}$. The only remaining step is to derive explicitly the asymptotic variance in the special case.

In this case, the optimal weighting matrix is given as:

$$W^*_{ij,k,t} = \frac{1}{\zeta_S}\left(\frac{S_j C_{it,k}}{\sigma_i^2} + \frac{S_i C_{jt,k}}{\sigma_j^2}\right) = \begin{cases} \dfrac{S_j}{\zeta_S\,\sigma_i^2}, & k = i, \\ \dfrac{S_i}{\zeta_S\,\sigma_j^2}, & k = j, \\ 0, & k \neq i, j. \end{cases}$$

Plugging this into $\mathbb{E}\left[\mathbf{W}_t^{*\prime}\mathbf{V}^{\psi}\mathbf{W}_t^{*}\right]$ with the expression for $\mathbf{V}^{\psi}$, we obtain the closed-form entries for the $N \times N$ matrix $\mathbb{E}\left[\mathbf{W}_t^{*\prime}\mathbf{V}^{\psi}\mathbf{W}_t^{*}\right]$. Define $\Sigma \equiv \sum_m S_m^2\sigma_m^2$. Then for any $k, \ell \in \{1, \ldots, N\}$,

$$\left[\,\mathbb{E}[\mathbf{W}_t^{*\prime}\mathbf{V}^{\psi}\mathbf{W}_t^{*}]\,\right]_{k\ell} = \frac{1}{\zeta_S^2}\begin{cases} \dfrac{\Sigma}{\sigma_k^2} - S_k^2, & k = \ell, \\ S_k S_\ell, & k \neq \ell. \end{cases} \tag{S4.7}$$

Inverting this matrix and scaling by ζ_S^2 gives the asymptotic variance in equation (3.4) of the main text.

S4.2 Proof of Proposition A.1

Proof. The proof follows Newey and McFadden (1994). The estimator of Appendix A.2 first regresses $q_{i,t}$ and $p_t\mathbf{C}'_{i,t}$ on $\mathbf{X}_{i,t}$ and then applies the optimal GIV to the residuals. This two-step estimator can be formulated as a joint GMM estimator that solves:

$$\frac{1}{T}\sum_t \mathbf{g}_t\left(\mathbf{b}, \mathbf{z}\right) \equiv \frac{1}{T}\sum_t \begin{bmatrix} \mathbf{g}_t^{OLS}\left(\mathbf{b}\right) \\ \mathbf{W}_t^{*\prime}\boldsymbol{\psi}_t(\mathbf{b}, \mathbf{z}) \end{bmatrix} = 0$$

where $\mathbf{g}^{OLS}\left(\mathbf{b}\right)$ is the moment condition of the first step, which identifies $(\boldsymbol{\beta}^q, \boldsymbol{\beta}^{pC})$, and the second block contains the optimal GIV moment conditions, taking the OLS coefficients as given.[10] As the coefficients are just-identified, the solutions to the GMM estimation are identical to the two-step estimator, and hence we can utilize properties of the GMM estimator to study the asymptotic behaviors. Use $\boldsymbol{\theta}$ to denote the vector of $(\boldsymbol{\beta}, \boldsymbol{\zeta})$ and $\hat{\boldsymbol{\theta}}$

[10] Here, we take the shortcut by formulating the GMM system in terms of $\boldsymbol{\beta}^q$ and $\boldsymbol{\beta}^{pC}$ rather than $\boldsymbol{\beta}$ directly as in (A.8). The derivation is slightly more complicated as the (A.8) involves both $\boldsymbol{\beta}$ and $\boldsymbol{\zeta}$ in the moment conditions, but the results are equivalent.

the corresponding estimator, we have:

$$\sqrt{T}\left(\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}\right)\xrightarrow{d}\mathcal{N}\left(\mathbf{0},\left(\mathbf{D}^{g\prime}\right)^{-1}\left(\mathbf{V}^{g}\right)\left(\mathbf{D}^{g}\right)^{-1}\right)$$
$$\mathbf{D}^{g}=\mathbb{E}\left[\frac{\partial\mathbf{g}_{t}\left(\boldsymbol{\theta}\right)}{\partial\boldsymbol{\theta}}\right]$$
$$\mathbf{V}^{g}=\mathbb{E}\left[\mathbf{g}_{t}\left(\boldsymbol{\theta}\right)\mathbf{g}_{t}\left(\boldsymbol{\theta}\right)'\right].$$

A crucial feature of these matrices is that they are block-diagonal:

$$\mathbf{D}^{g}=\begin{bmatrix}\frac{\partial\mathbf{g}_{t}^{OLS}(\mathbf{b})}{\partial\mathbf{b}}\mid_{\mathbf{b}=\boldsymbol{\beta}} & \\ & \mathbb{E}\left[\mathbf{W}_{t}^{*\prime}\mathbf{D}_{t}\right]\end{bmatrix}$$
$$\mathbf{V}^{g}=\begin{bmatrix}\mathbf{V}^{OLS} & \mathbf{0}\\ \mathbf{0} & \mathbf{W}^{*\prime}\mathbf{V}^{\psi}\mathbf{W}^{*}\end{bmatrix}.$$

To see $\mathbf{D}^{g}$ is diagonal, notice that the estimate for $\boldsymbol{\zeta}$ does not enter the moment conditions for the OLS step by the two-step nature, and

$$\mathbb{E}\left[\frac{\partial\psi_{ij,t}\left(\boldsymbol{b},\boldsymbol{\zeta}\right)}{\partial\mathbf{b}}\mid_{\mathbf{b}=\boldsymbol{\beta}}\right]=\mathbb{E}\left[\frac{\partial\left(u_{it}-\mathbf{X}_{it}'\left(\mathbf{b}-\boldsymbol{\beta}\right)\right)\left(u_{jt}-\mathbf{X}_{jt}'\left(\mathbf{b}-\boldsymbol{\beta}\right)\right)}{\partial\mathbf{b}}\mid_{\mathbf{b}=\boldsymbol{\beta}}\right]=\mathbf{0}.$$

$\mathbf{V}^{g}$ being diagonal is a result that the error term in $\mathbf{g}_{t}^{OLS}$ is linear in u_{it}, while the error terms of $\boldsymbol{\psi}_{t}$ are quadratic $u_{it}u_{jt}$ for pairs $i\neq j$. By the cross-sectional independence assumption (Assumption A.1 part 3), any linear combination between them has a covariance of zero.

Due to the block-diagonal structure in $\mathbf{D}^{g}$ and $\mathbf{V}^{g}$, the asymptotic variance of $\hat{\boldsymbol{\zeta}}$ is not affected by the first step estimation.

To further derive the asymptotic variance of $\hat{\boldsymbol{\beta}}$, we use the Delta method:

$$Var(\hat{\boldsymbol{\beta}})=\begin{bmatrix}\mathbf{I} & \boldsymbol{\zeta}\end{bmatrix}'Var(\begin{bmatrix}\hat{\boldsymbol{\beta}}^{q} & \hat{\boldsymbol{\beta}}^{pC}\end{bmatrix})\begin{bmatrix}\mathbf{I} & \boldsymbol{\zeta}\end{bmatrix}'+\boldsymbol{\beta}^{pC}Var(\hat{\boldsymbol{\zeta}})\boldsymbol{\beta}^{pC\prime}$$

where the first part is equivalent to asymptotic variance of the OLS estimator of regressing $q_{i,t}+p_{t}\mathbf{C}_{i,t}'\boldsymbol{\zeta}$ on $\mathbf{X}_{i,t}$, which is given in (A.10). □

S4.3 Proof of Proposition A.2

Proof. Define $u^b_{i,t} := \Delta q_{i,t} + \zeta^b_i \Delta p_t$ as the misspecified residuals. Using the true model in (A.11), it can be expressed as:

$$u^b_{i,t} = -\left(\zeta_i - \zeta^b_i\right)\Delta p_t + u_{i,t} \tag{S4.8}$$

Plugging Equation (A.12) into the misspecified residual, and denoting $b_i := \frac{\zeta_i - \zeta^b_i}{\zeta_S}$, we have:

$$u^b_{i,t} = -b_i u_{S,t} + u_{i,t} \tag{S4.9}$$

Aggregating across sectors with size weights S_i, define $u^b_{S,t} \equiv \sum_i S_i u^b_{i,t}$, we obtain:

$$u^b_{S,t} = (1 - b_S)u_{S,t}$$

Here $b_S \equiv \sum_i S_i\, b_i$ denotes the size-weighted average of the sectoral coefficients b_i.

The optimal GIV estimator under the misspecified model solves:

$$\begin{aligned}\mathbb{E}\left[u^b_{i,t}\sum_{j\neq i} S_j u^b_{j,t}\right] &= \mathbb{E}\left[u^b_{i,t}\left(u^b_{S,t} - S_i u^b_{i,t}\right)\right], \\ &= \mathbb{E}\Big[\big(-b_i u_{S,t} + u_{i,t}\big)\big((1 - b_S + S_i b_i)u_{S,t} - S_i u_{i,t}\big)\Big]\end{aligned} \tag{S4.10}$$

Express it in terms of the moments of shocks:

$$-b_i(1 - b_S + S_i b_i)\sigma^2_{u,S} + (1 - b_S + 2S_i b_i)\sigma_{i,S} - S_i\sigma^2_{u,i} = 0 \tag{S4.11}$$

where

$$\begin{aligned}\sigma^2_{u,S} &:= \mathbb{E}[u^2_{S,t}] = \mathbb{E}\Big[(\lambda_S\eta_t + \sum_j S_j\epsilon_{j,t})^2\Big] = \lambda^2_S\sigma^2_\eta + \sum_j S^2_j\sigma^2_{\epsilon,j}, \\ \sigma_{i,S} &:= \mathbb{E}[u_{i,t}u_{S,t}] = \mathbb{E}\Big[(\lambda_i\eta_t + \epsilon_{i,t})(\lambda_S\eta_t + \sum_j S_j\epsilon_{j,t})\Big] = \lambda_i\lambda_S\sigma^2_\eta + S_i\sigma^2_{\epsilon,i}, \\ \sigma^2_{u,i} &:= \mathbb{E}[u^2_{i,t}] = \mathbb{E}\Big[(\lambda_i\eta_t + \epsilon_{i,t})^2\Big] = \lambda^2_i\sigma^2_\eta + \sigma^2_{\epsilon,i},\end{aligned}$$

using $\mathbb{E}[\eta_t\epsilon_{i,t}] = 0$ and $\mathbb{E}[\epsilon_{i,t}\epsilon_{j,t}] = 0$ for $i \neq j$. Equation (S4.11) gives N quadratic equations

in b_i.

Symmetric benchmark. To proceed, we first construct a symmetric benchmark. Use upper bar to denote variables for the symmetric benchmark:

$$\bar{S}_i := \frac{1}{N}, \qquad \bar{\lambda}_i := \lambda_S, \qquad \bar{\sigma}^2_{\epsilon,i} = \frac{\sum_i S_i^2 \sigma^2_{\epsilon,i}}{\sum_i \bar{S}_i^2} \quad \forall i.$$

Notice that by construction, this symmetric benchmark has the same variance of the total demand shock $\bar{\sigma}^2_{u,S}$ and aggregate loading $\bar{\lambda}_S$, and hence the same $\mathcal{R}^2$ as the heterogeneous model.

Using the symmetry, we have:

$$\bar{\sigma}^2_{u,S} = \sigma^2_{u,S}, \qquad \bar{\sigma}_{i,S} = \sigma^2_{u,S}, \qquad \bar{\sigma}^2_{u,i} = \lambda_S^2 \sigma_\eta^2 + \bar{\sigma}^2_\epsilon = \lambda_S^2 \sigma_\eta^2 + N \sigma^2_{\epsilon,S}.$$

In this case, the equation systems in (S4.11) are symmetric, and hence $\bar{b}_i = \bar{b}_S = \bar{b}$. Plugging in and simplifying, we have:

$$\bar{b}^2 - 2\bar{b} + \mathcal{R}^2 = 0 \tag{S4.12}$$

where $\mathcal{R}^2 \equiv \frac{\lambda_S^2 \sigma_\eta^2}{\sigma^2_{u,S}}$. Focusing on the root $\bar{b} < 1$ so $\bar{\zeta}^b$ remains positive, we have $1 - \bar{b} = \sqrt{1 - \mathcal{R}^2}$, hence

$$\zeta_S^b = \zeta_S \sqrt{1 - \mathcal{R}^2}.$$

which is the exact symmetric-case formula in Equation (A.15).

First-order perturbation. Now we consider the first-order perturbation around the symmetric benchmark. Define the deviations from the symmetric benchmark as:

$$\Delta b_i = b_i - \bar{b}, \qquad \Delta b_S = b_S - \bar{b}, \qquad \omega_i = \frac{S_i - \bar{S}}{\bar{S}}, \qquad \delta_i = \frac{\lambda_i - \bar{\lambda}}{\bar{\lambda}}, \qquad \kappa_i = \frac{\sigma^2_{\epsilon,i} - \bar{\sigma}^2_\epsilon}{\bar{\sigma}^2_\epsilon},$$

We can express Equation (S4.11) as a function of deviations from the symmetric benchmark:

$$F_i(\Delta b_i, \Delta b_S, \omega_i, \delta_i, \kappa_i \mid \sigma^2_{u,S}, \lambda_S) = 0.$$

We fix $\sigma^2_{u,S}$ and λ_S because the perturbation in heterogeneity $(\omega_i, \delta_i, \kappa_i)$ does not enter λ_S and $\sigma^2_{u,S}$ to the first order.

The first order approximation of F_i around the symmetric benchmark is then given

by:

$$\partial_b F_i \Delta b_i + \partial_{b_S} F_i \Delta b_S + \partial_\omega F_i \omega_i + \partial_\delta F_i \delta_i + \partial_\kappa F_i \kappa_i + O(\xi^2) = 0.$$

We make the following claim, and prove it at the end of this section.

Claim 1. With $\bar{S} = 1/N$ and $\omega_i, \delta_i, \kappa_i$ as defined,

$$\sum_i S_i \Delta b_i = \Delta b_S, \qquad \sum_i S_i \omega_i = O(\xi^2), \qquad \sum_i S_i \delta_i = 0, \qquad \sum_i S_i \kappa_i = O(\xi^2).$$

With this claim, the last three terms vanish in the first order in the size-weighted summation of the first-order perturbation, which gives:

$$(\partial_b F_i + \partial_{b_S} F_i)\ \Delta b_S + O(\xi^2) = 0 \tag{S4.13}$$

As we show below, $\partial_b F_i + \partial_{b_S} F_i \neq 0$. Hence, Equation (S4.13) implies that to the first order in heterogeneity, the bias in aggregate elasticity is the same as the symmetric benchmark:

$$\Delta b_S = O(\xi^2) \implies b_S = \bar{b} + O(\xi^2)$$

Individual bias around the symmetric benchmark. To derive the first-order approximation of the bias for individual elasticity, we evaluate the partial derivatives of F_i around the symmetric benchmark.

As an intermediate step, we first linearize the moments:

$$\begin{aligned}
\sigma_{u,S}^2 &= \bar{\sigma}_{u,S}^2 + O(\xi^2) \quad \text{(no first-order change)}, \\
\sigma_{i,S} &= \bar{\sigma}_{u,S}^2 + \bar{\lambda}^2 \sigma_\eta^2 \delta_i + \bar{S} \bar{\sigma}_\epsilon^2 (\omega_i + \kappa_i) + O(\xi^2), \\
\sigma_{u,i}^2 &= \bar{\lambda}^2 \sigma_\eta^2 + \bar{\sigma}_\epsilon^2 + 2\bar{\lambda}^2 \sigma_\eta^2 \delta_i + \bar{\sigma}_\epsilon^2 \kappa_i + O(\xi^2).
\end{aligned}$$

With these quantities, we can derive the partial derivatives of F_i evaluated at the sym-

metric benchmark:

$$\begin{aligned}
\frac{\partial F_i}{\partial \Delta b_i} &= \sigma_{u,S}^2 (1-\bar{b})(2\bar{S}-1), \\
\frac{\partial F_i}{\partial \Delta b_S} &= -\sigma_{u,S}^2 (1-\bar{b}), \\
\frac{\partial F_i}{\partial \omega_i} &= \sigma_{u,S}^2 (1-\bar{b})^2 \bar{b} (2\bar{S}-1), \\
\frac{\partial F_i}{\partial \delta_i} &= -\lambda_S^2 \sigma_\eta^2 (1-\bar{b})(2\bar{S}-1) = -\sigma_{u,S}^2 \mathcal{R}^2 (1-\bar{b})(2\bar{S}-1), \\
\frac{\partial F_i}{\partial \kappa_i} &= \sigma_{u,S}^2 (1-\bar{b})^2 \bar{b} (2\bar{S}-1).
\end{aligned} \tag{S4.14}$$

The derivatives with respect to ω_i and κ_i coincide because, to first order, both only scale $S_i \sigma_{\epsilon,i}^2$. Linearizing the moments gives $\partial \sigma_{i,S}/\partial \omega_i = \partial \sigma_{i,S}/\partial \kappa_i = \bar{S}\bar{\sigma}_\epsilon^2$, $\partial \sigma_{u,i}^2/\partial \omega_i = 0$, $\partial \sigma_{u,i}^2/\partial \kappa_i = \bar{\sigma}_\epsilon^2$, and $\partial S_i/\partial \omega_i = \bar{S}$; evaluating F_i at the symmetric benchmark and using $\bar{b}^2 - 2\bar{b} + \mathcal{R}^2 = 0$ reduces both derivatives to $\bar{S}\bar{\sigma}_\epsilon^2 \bar{b}(2\bar{S}-1)$, which equals $\sigma_{u,S}^2 (1-\bar{b})^2 \bar{b}(2\bar{S}-1)$ because $\bar{S}\bar{\sigma}_\epsilon^2 = \sigma_{u,S}^2(1-\mathcal{R}^2)$ and $1-\mathcal{R}^2 = (1-\bar{b})^2$. Plugging in the partial derivatives and $\Delta b_S = O(\xi^2)$, we solve for Δb_i as:

$$\Delta b_i = \mathcal{R}^2 \delta_i - b\big(1-b\big)(\omega_i + \kappa_i) + O(\xi^2).$$

where we use $b = \bar{b}$ for simplicity. Or equivalently,

$$\frac{\zeta_i - \zeta_i^b}{\zeta_S} = b + \mathcal{R}^2 \delta_i - b\big(1-b\big)(\omega_i + \kappa_i) + O(\xi^2).$$

□

Proof of Claim 1. (i) follows from $b_S = \sum_i S_i b_i$. (ii) $\sum_i \omega_i = 0$ by construction, so $\sum_i S_i \omega_i = \bar{S} \sum_i (1+\omega_i)\omega_i = \bar{S} \sum_i \omega_i^2 = O(\xi^2)$. (iii) $\sum_i S_i \delta_i = \lambda_S^{-1} \sum_i S_i \lambda_i - \sum_i S_i = 0$. (iv) The definition of $\bar{\sigma}_\epsilon^2$ gives $\sum_i S_i^2 \kappa_i = \sum_i \bar{S}^2 - \sum_i S_i^2 = -\bar{S}^2 \sum_i \omega_i^2 = O(\xi^2)$. Writing $S_i = S_i^2/\bar{S} - \bar{S}(\omega_i + \omega_i^2)$ then gives $\sum_i S_i \kappa_i = \bar{S}^{-1} \sum_i S_i^2 \kappa_i - \bar{S} \sum_i (\omega_i + \omega_i^2)\kappa_i$; the first term is $O(\xi^2)$, and so is the second by Cauchy–Schwarz, since $\sum_i (\omega_i + \omega_i^2)^2 = O(\xi^2)$ and $\sum_i \kappa_i^2 \le C(N) \sum_i S_i^2 \kappa_i^2 \le C(N)\xi^2$ for fixed N. □

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